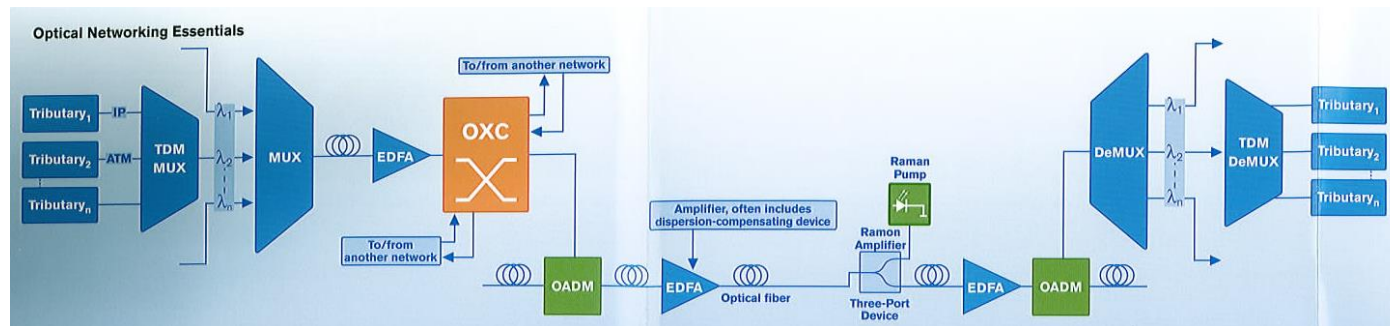


Module Ouverture Recherche Photonique et communications Optiques (ORPHeO)

“INTRODUCTION TO FIBER-OPTIC COMMUNICATIONS”



Summary



Preamble

Objectives

Why fiber optic is a so successful technology ?

Main features of the fiber optic

Fiber optic based communication systems: several generations

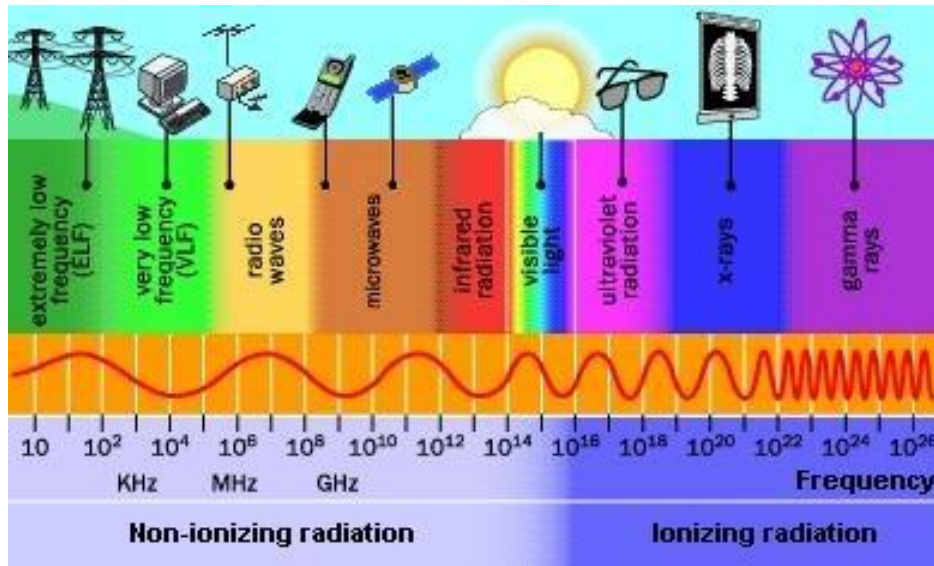
Fiber optic communication systems: limit of capacity ?

Fiber optic system: how to calculate the link power budget ?

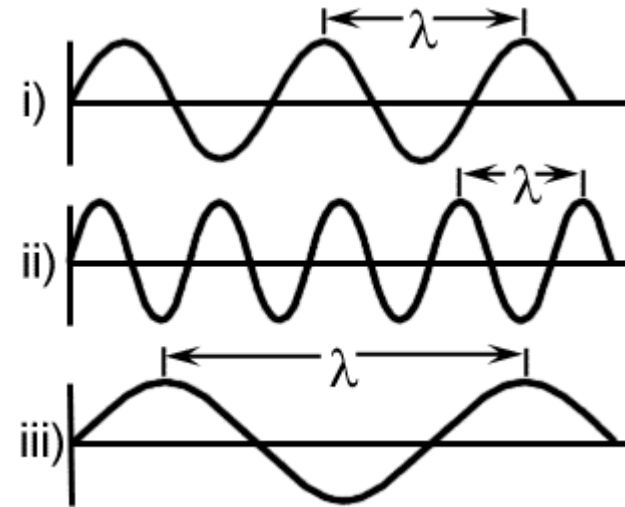
- S.I. Units

10^{-15}	femto	f	10^{+3}	kilo	k
10^{-12}	pico	p	10^{+6}	mega	M
10^{-9}	nano	n	10^{+9}	giga	G
10^{-6}	micro	μ	10^{+12}	tera	T
10^{-3}	milli	m	10^{+15}	peta	P

- The Optical Spectrum



- The Optical Wave

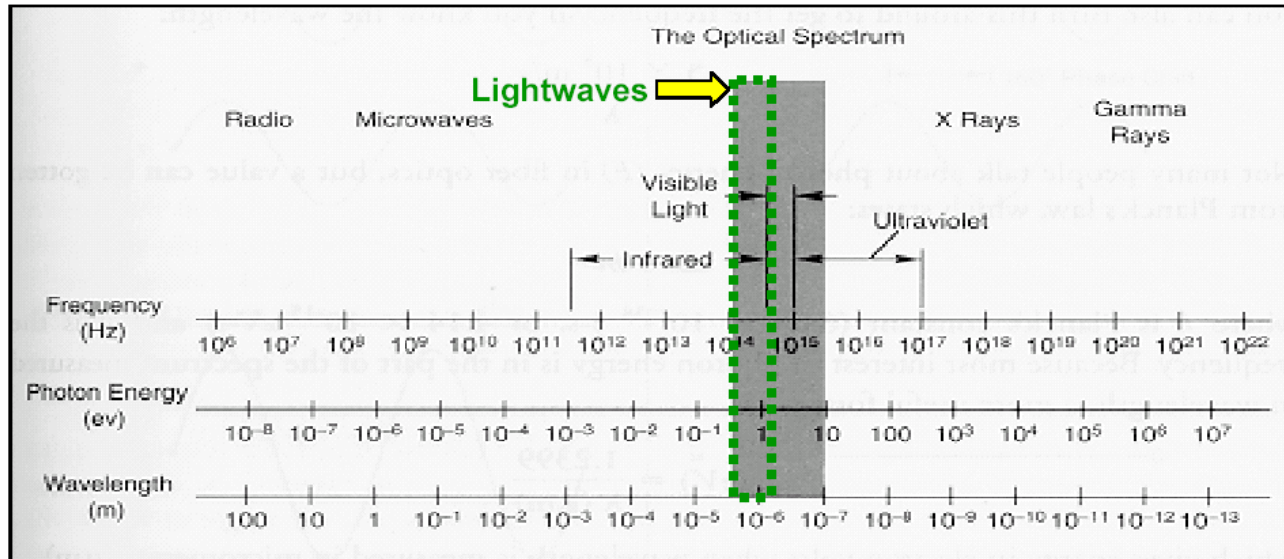


Wavelength: λ in m (typically μm or nm)

Optical frequency : ν ($= c/\lambda$) in Hz (typically THz)

c : speed of light in vacuum: $3 \cdot 10^8 \text{ m.s}^{-1}$

- The Optical Spectrum



Objectives

- To provide an overview of modern optical fibre communication systems
- To understand why this technology is so successful
- To understand the underlying physical limitations, how they have been addressed over the years, and ... what remains to be solved

When you finish this module, you will be able to:

- Identify the basic components of a fiber optic communication system
- Discuss light propagation in an optical fiber
- Identify the various types of optical fibers
- Determine the dispersion characteristics for the various types of optical fibersDescribe the various connector types
- Calculate decibel and dBm power
- Calculate the power budget for a fiber optic system
- Calculate the bandwidth of a fiber optic system

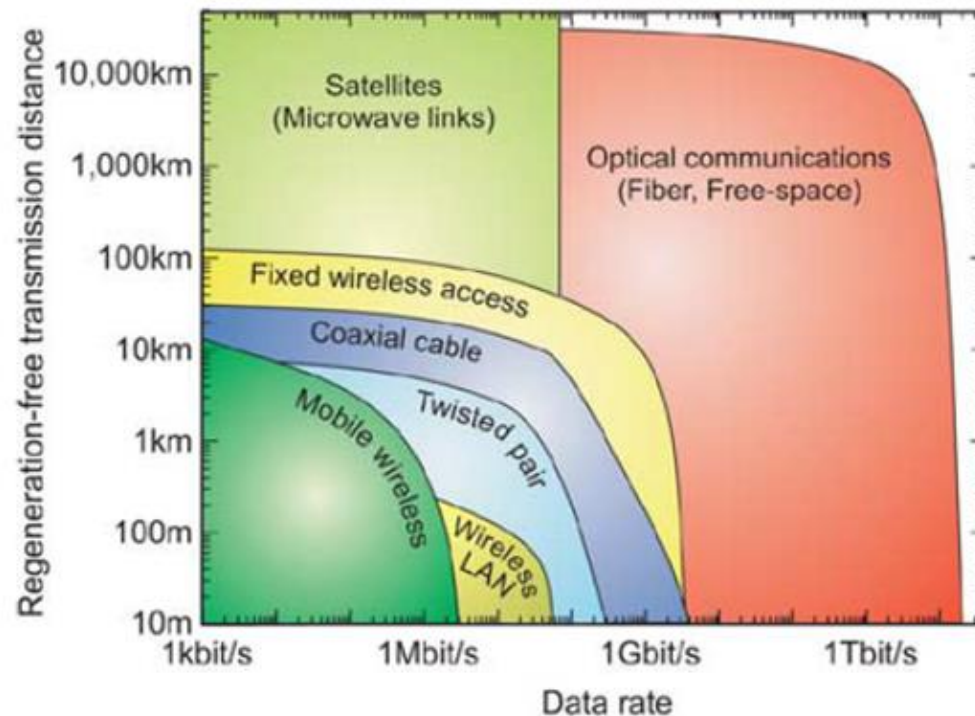
Why fiber optic is a so successful technology ?

Why fiber optic is a so successful technology ?

Bandwidth is a compelling reason

1970's – Copper cable, 672 simultaneous data streams, with 2 km spacing between amplification points

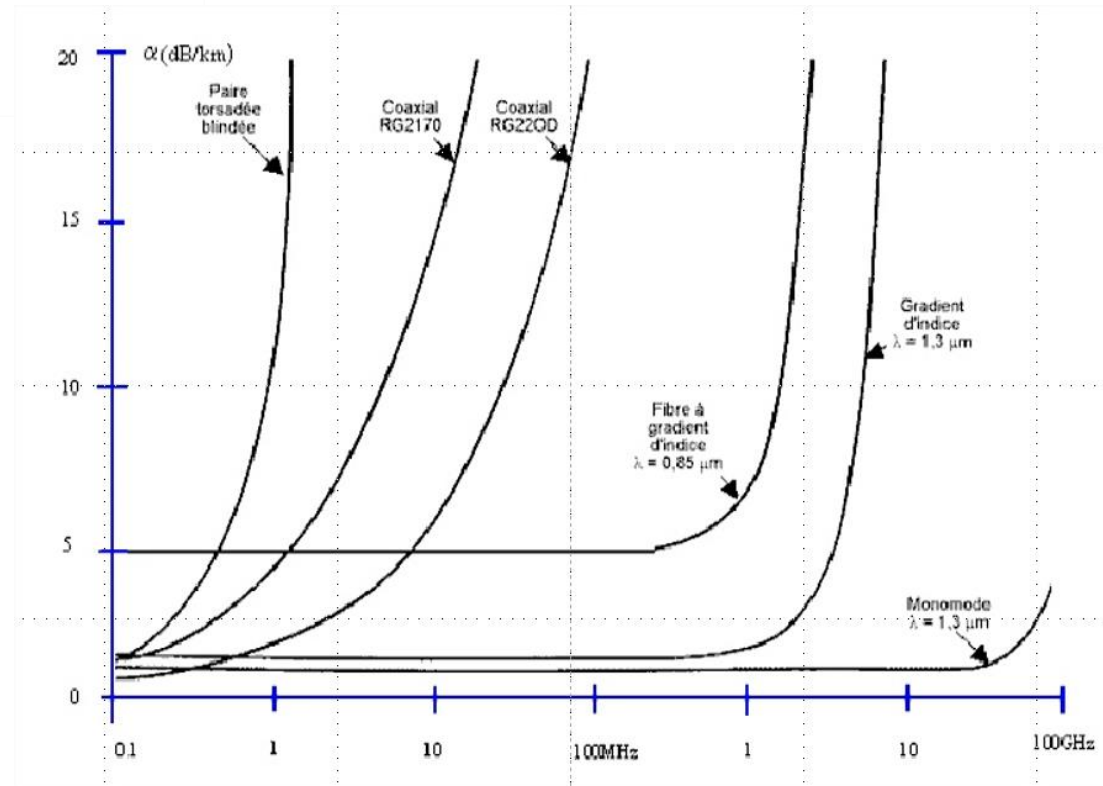
Today – With a single fiber, in excess of 130,000 simultaneous data streams, with 60 km spacing between amplification points



By PETER J. WINZER, *Senior Member IEEE*, AND RENÉ-JEAN ESSIAMBRE, *Member IEEE*

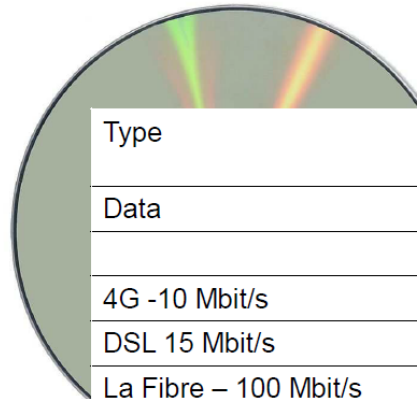
Why fiber optic is a so successful technology ?

- Low loss - Flat attenuation (loss) across frequency
- Electromagnetic immunity - No radiation or absorption of electromagnetic energy
- Light weight - Copper co-ax cable can weigh 9 times as much as fiber cable
- Small size - A smaller diameter fiber cable provides more bandwidth than a larger copper cable
- Safety - No electrical hazard, no spark potential
- Security - Extremely difficult to “fiber tap”



Why fiber optic is a so successful technology ?

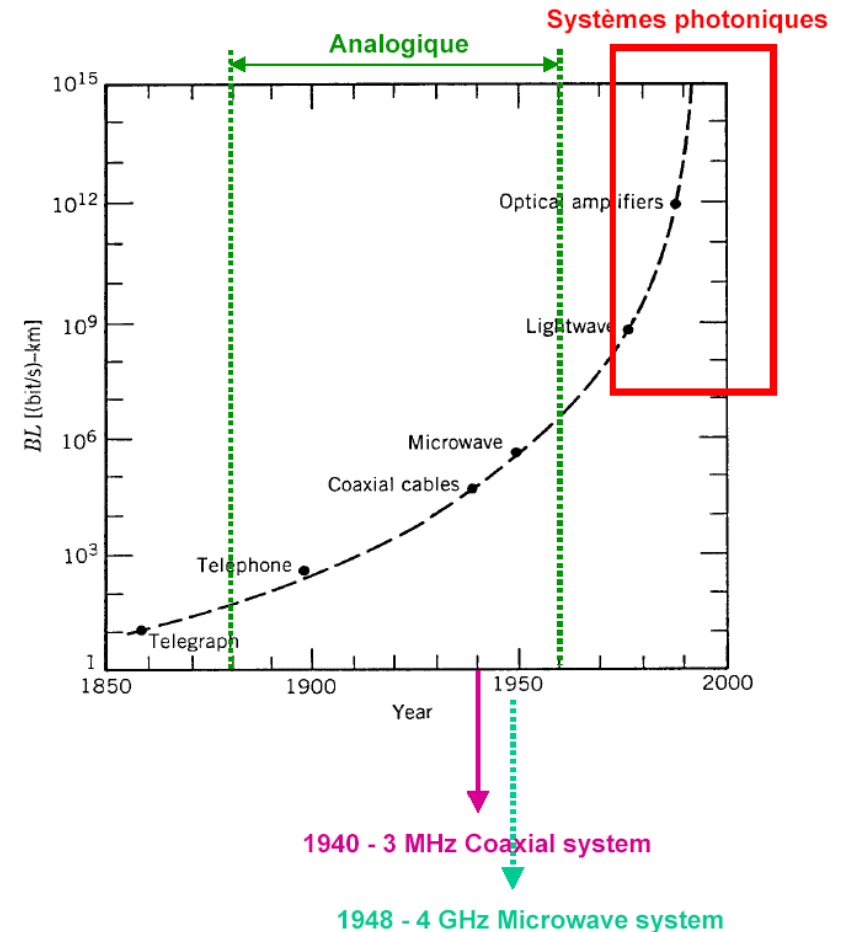
Transporting data

		
	Single-sided double layer	Dual layer
Type		
Data	8.5 Gbytes	50 Gbytes
4G - 10 Mbit/s	~2 h	~11 h
DSL 15 Mbit/s		
La Fibre – 100 Mbit/s	~ 12 min	~ 1 h
Core 10 Gbit/s (1995)	6.8 s	40 s
Core 100 Gbit/s (2010)	0.68 s	4 s
Readout	177.28 Mbit/s (16x)	288 Mbit/s (8x – USB29)

transmission capacity of fiber

bandwidth × distance

Huge Binary rate-Length (BL in bit.s⁻¹.km) evolution:



Why fiber optic is a so successful technology ?

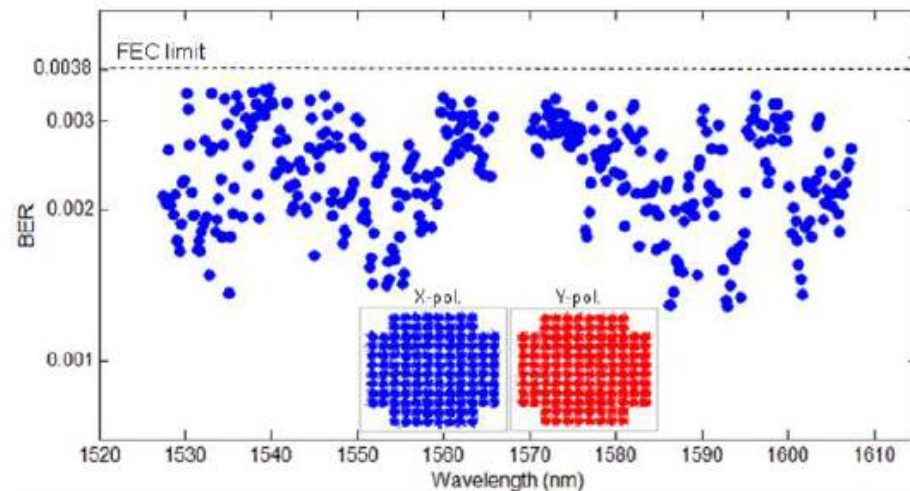
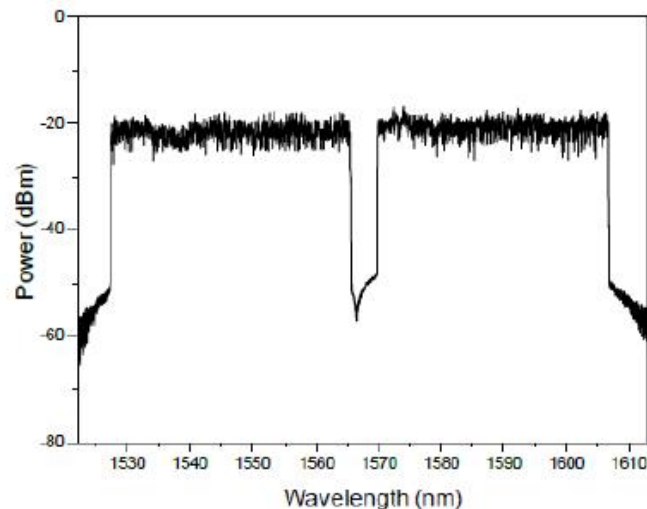
Record capacity over conventional fibres

101.7-Tb/s (370×294 -Gb/s) PDM-128QAM-OFDM

Transmission over 3×55 -km SSMF using Pilot-based Phase Noise Mitigation

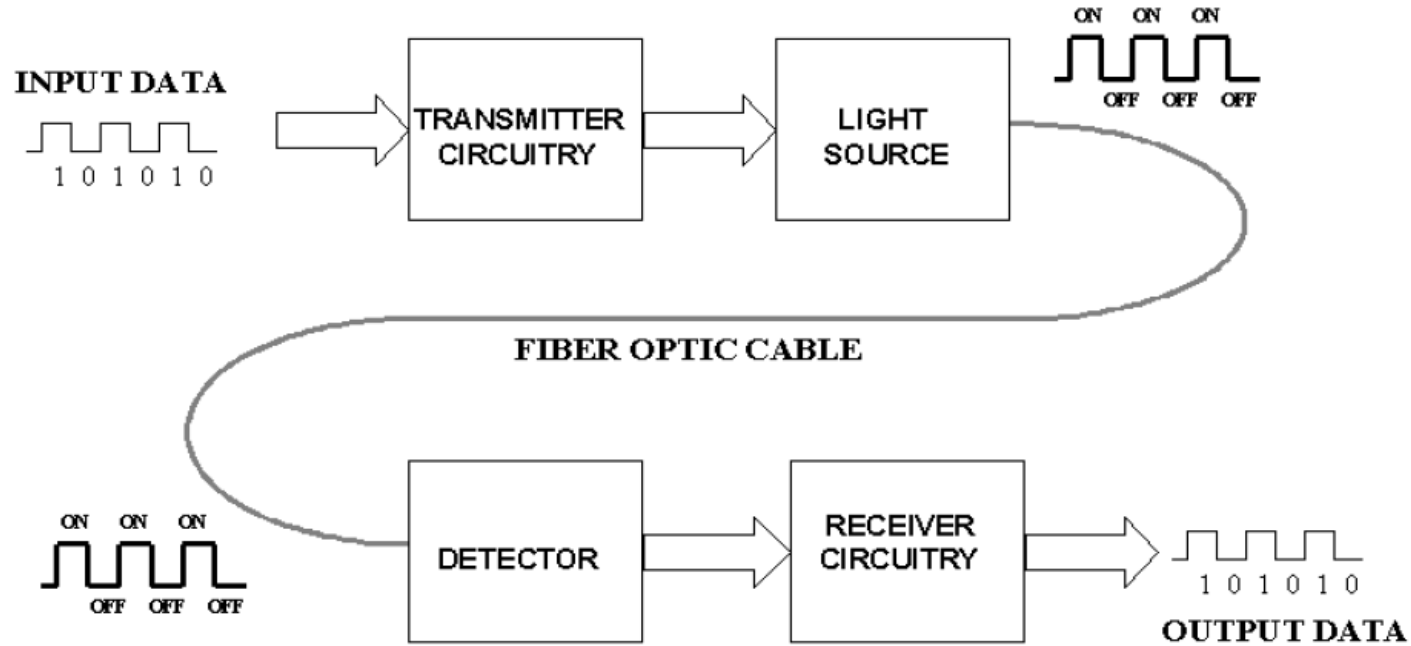
Dayou Qian, Ming-Fang Huang, Ezra Ip, Yue-Kai Huang, Yin Shao, Junqiang Hu, Ting Wang
NEC Laboratories America, Inc., 4 Independence Way, Princeton NJ 08540 USA
dqian@nec-labs.com

Abstract: Record capacity transmission of 101.7-Tb/s (370×294 -Gb/s) is performed over 3×55 km SSMF using PDM-128QAM-OFDM modulation and pilot-based phase noise mitigation. Achieved spectral efficiency of 11 bits/s/Hz is the highest reported to date for WDM transmission.

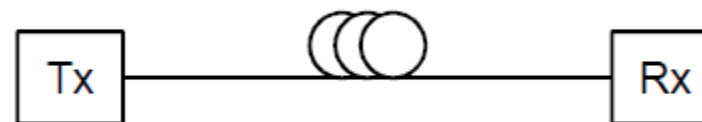




Fiber optic optical communication system: principle



Basic fiber optic communication system



v1.0

Main features of the fiber optic transmission channel

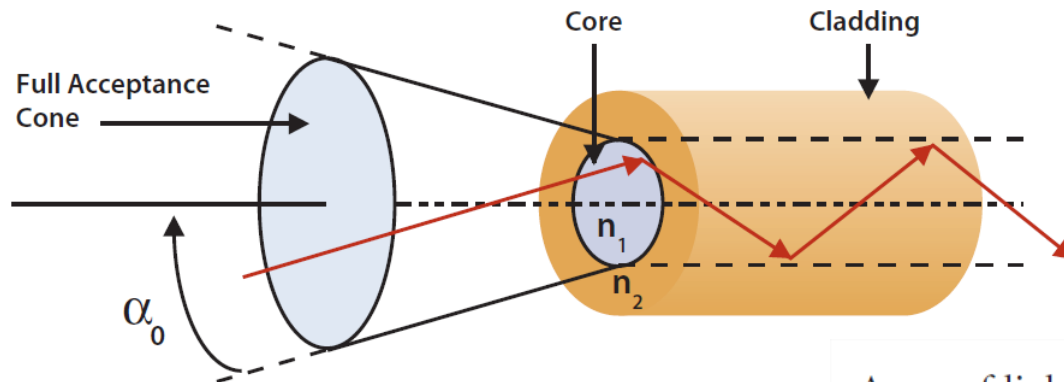
Back to basis, why is fiber an optical waveguide?

Fiber Design

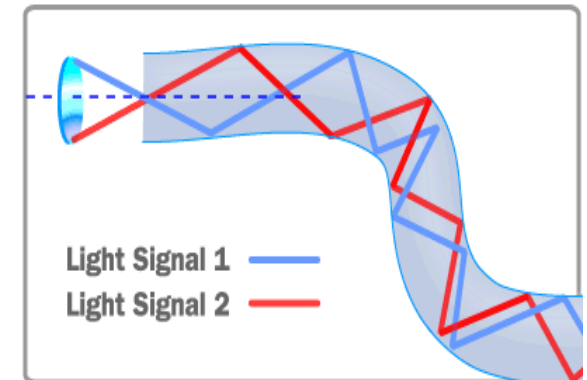


The composition of optical fiber

Transmission Principles



The injection of light into a fiber



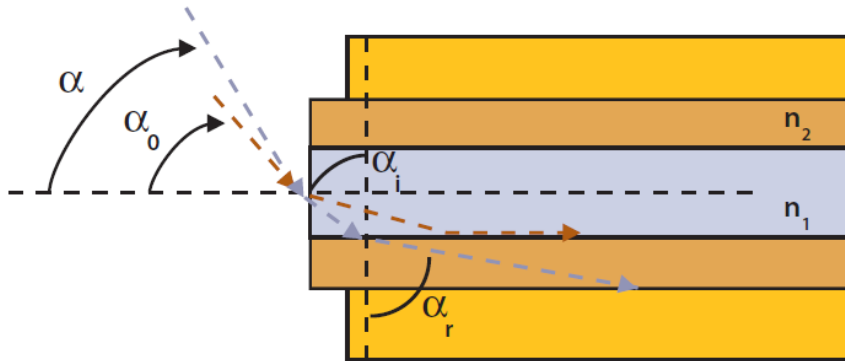
A ray of light enters a fiber at a small angle α . The capability (maximum acceptable value) of the fiber cable to receive light through its core is determined by its numerical aperture (NA).

$$NA = \sin \alpha_0 = \sqrt{n_1^2 - n_2^2}$$

Back to basis, why is fiber an optical waveguide?

Refraction

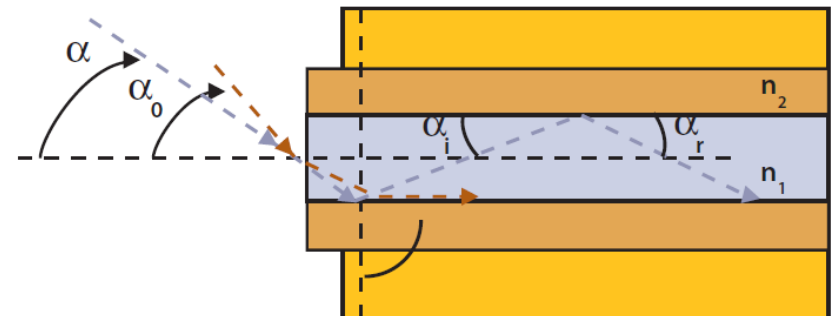
$$n_1 \sin \alpha_i = n_2 \sin \alpha_r$$



Refraction of light

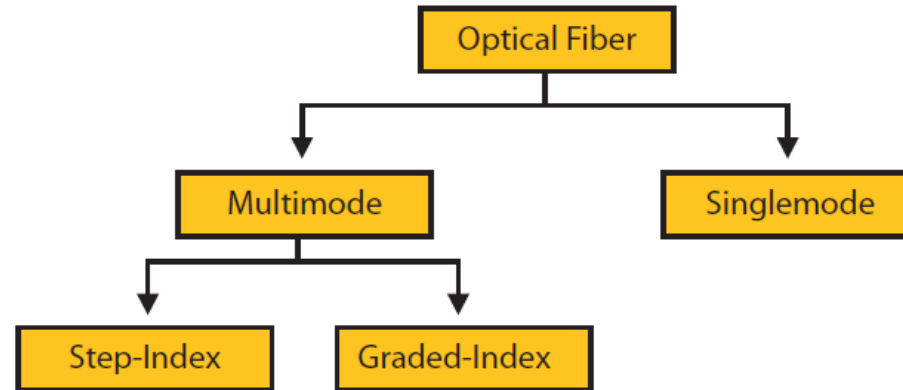
Reflection

$$\alpha_i = \alpha_r$$



Reflection of light

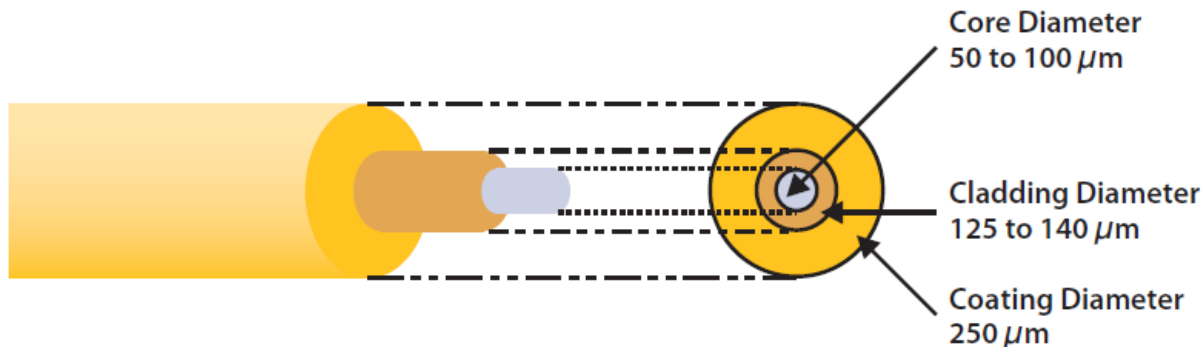
Fiber optic: multimode or singlemode?



Types of glass fiber

The primary advantages of multimode fiber are the ease of coupling to light sources and to other fiber, lower cost light sources (transmitters), and simplified connectorization and splicing processes. However, its relatively high attenuation and low bandwidth limit the transmission of light over multimode fiber to short distances.

Multimode Fiber

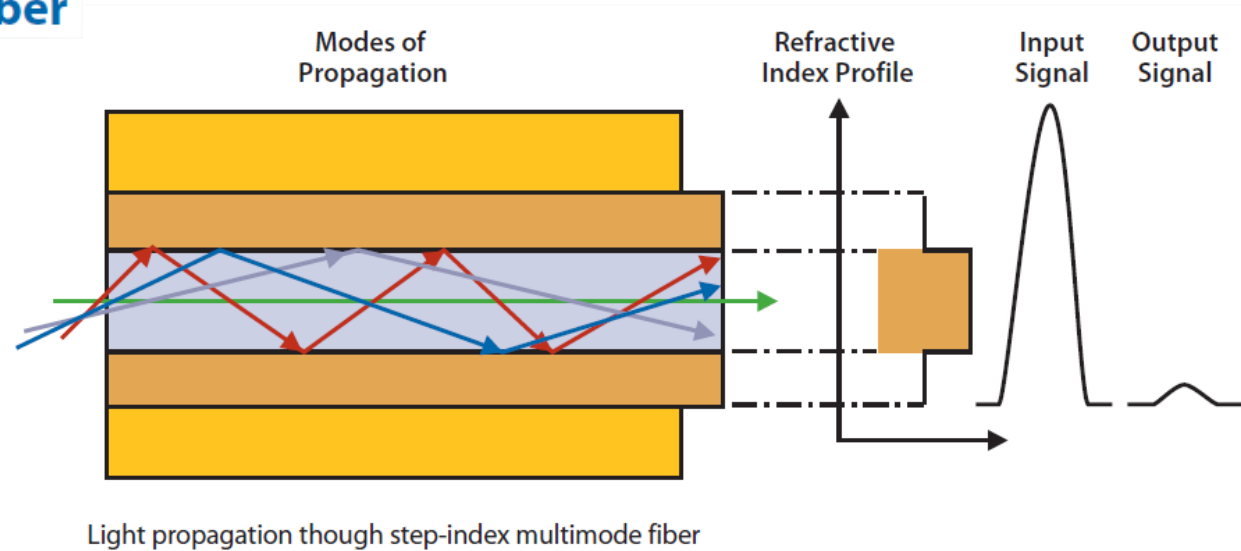


The composition of multimode fiber

Fiber optic: multimode or singlemode?

Step-Index Multimode Fiber

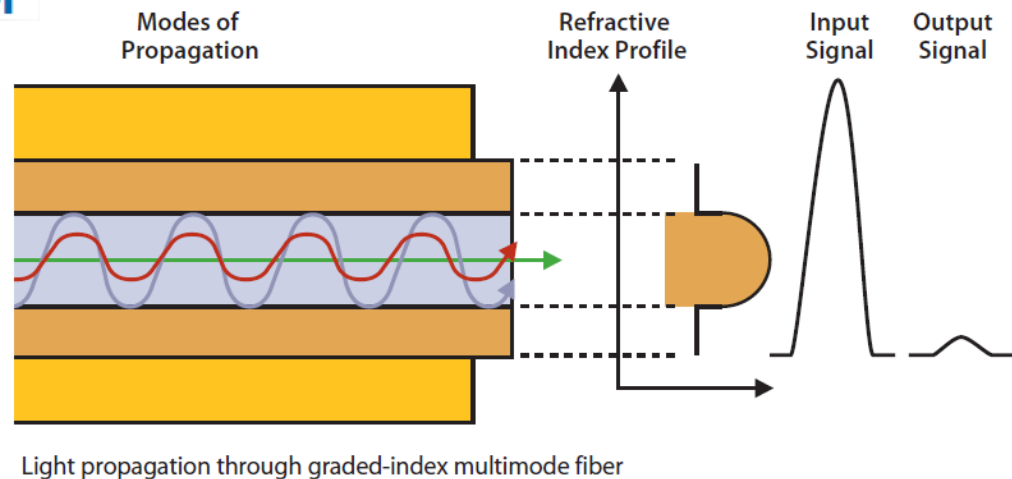
20 MHz·km



Graded-Index Multimode Fiber

3 dB/km at 850 nm
1 dB/km at 1300 nm

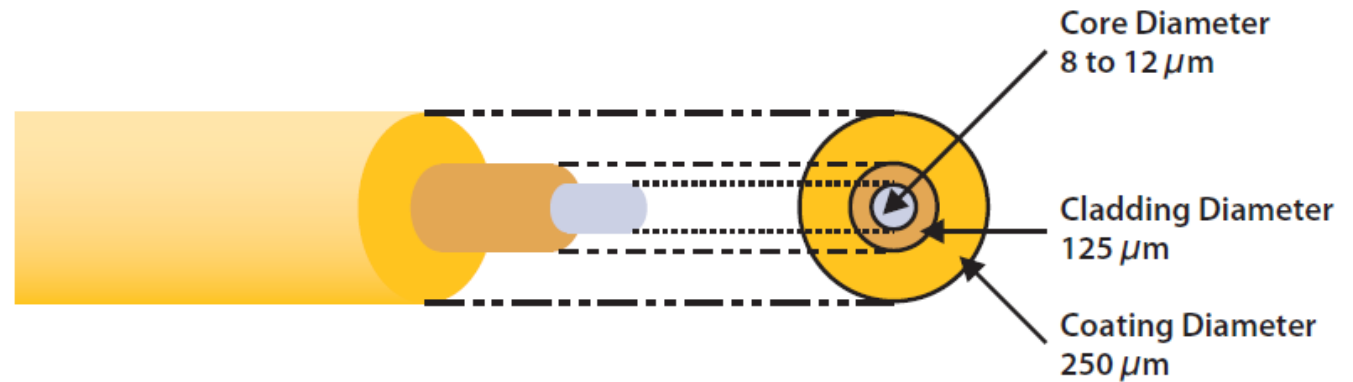
160 MHz·km at 850 nm
500 MHz·km at 1300 nm



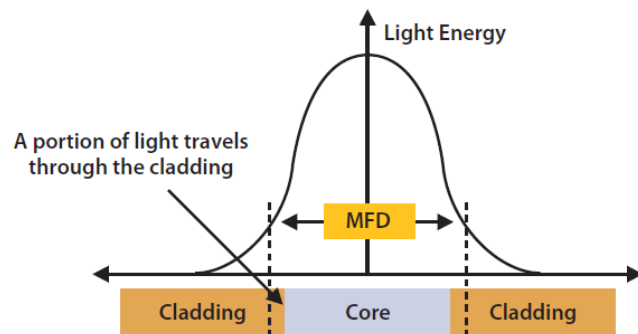
Fiber optic: multimode or singlemode?

Singlemode Fiber

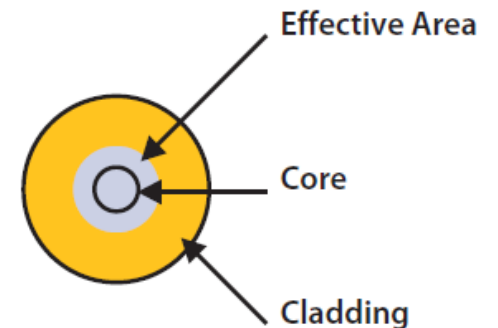
The typical dimensions of singlemode fiber range from a core of 8 to 12 μm and a cladding of 125 μm . The typical core-cladding angle is 8.5° . The refractive index of singlemode fiber is typically 1.465.



The composition of singlemode fiber



The mode field diameter (MFD) of singlemode fiber



The effective area of singlemode fiber

Fiber optic: multimode or singlemode?

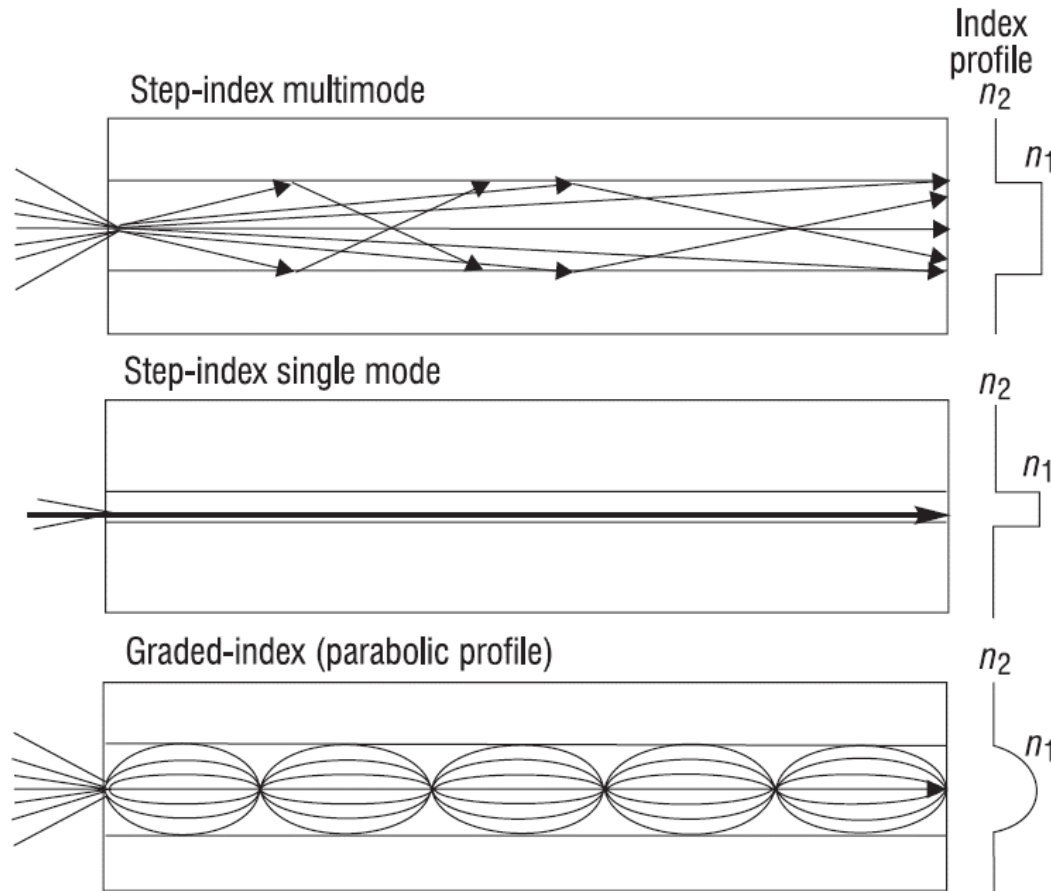
Singlemode Fiber

G.652: Characteristics of singlemode optical fiber and cable

	Characteristics	Wavelength Coverage	Applications
G.652.A	Max PMD = 0.5 $\text{ps}/\sqrt{\text{km}}$	1310 nm and 1550 nm regions (O and C bands)	Supports applications such as those recommended in G.957 and G.691 up to STM-16, 10 Gb/s up to 40 km (Ethernet), and STM-256 for G.693.
G.652.B	Maximum attenuation specified at 1625 nm. Max PMD = 0.2 $\text{ps}/\sqrt{\text{km}}$	1310 nm, 1550 nm, and 1625 nm regions (O and C+L bands)	Supports some higher bit rate applications up to STM-64 in G.691 and G.692 and some STM-256 applications in G.693 and G.959.1. Depending on the application, chromatic dispersion accommodation may be necessary.
G.652.C	Maximum attenuation specified at 1383 nm (equal or lower than 1310 nm). Max PMD = 0.5 $\text{ps}/\sqrt{\text{km}}$	From O to C bands	Similar to G.652.A, but this standard allows for transmission in portions of an extended wavelength range from 1360 nm to 1530 nm. Suitable for CWDM systems.
G.652.D	Maximum attenuation specified from 1310 to 1625 nm. Maximum attenuation specified at 1383 nm (equal or lower than 1310 nm). Max PMD = 0.2 $\text{ps}/\sqrt{\text{km}}$	Wide band coverage (from O to L bands)	Similar to G.652.B, but this standard allows for transmission in portions of an extended wavelength range from 1360 nm to 1530 nm. Suitable for CWDM systems.

Fiber optic: multimode or singlemode?

Review of Singlemode and Multimode Fiber



Comments

- Large N.A.
- Easy coupling
- Modal dispersion
- Lower data rates
- Shorter distances

- Small N.A.
- Coupling more difficult
- No modal dispersion
- High data rates
- Long distances

- Large N.A.
- Easy coupling
- Less modal dispersion
- Good compromise between multimode and single-mode fiber

Fiber optic: multimode or singlemode?

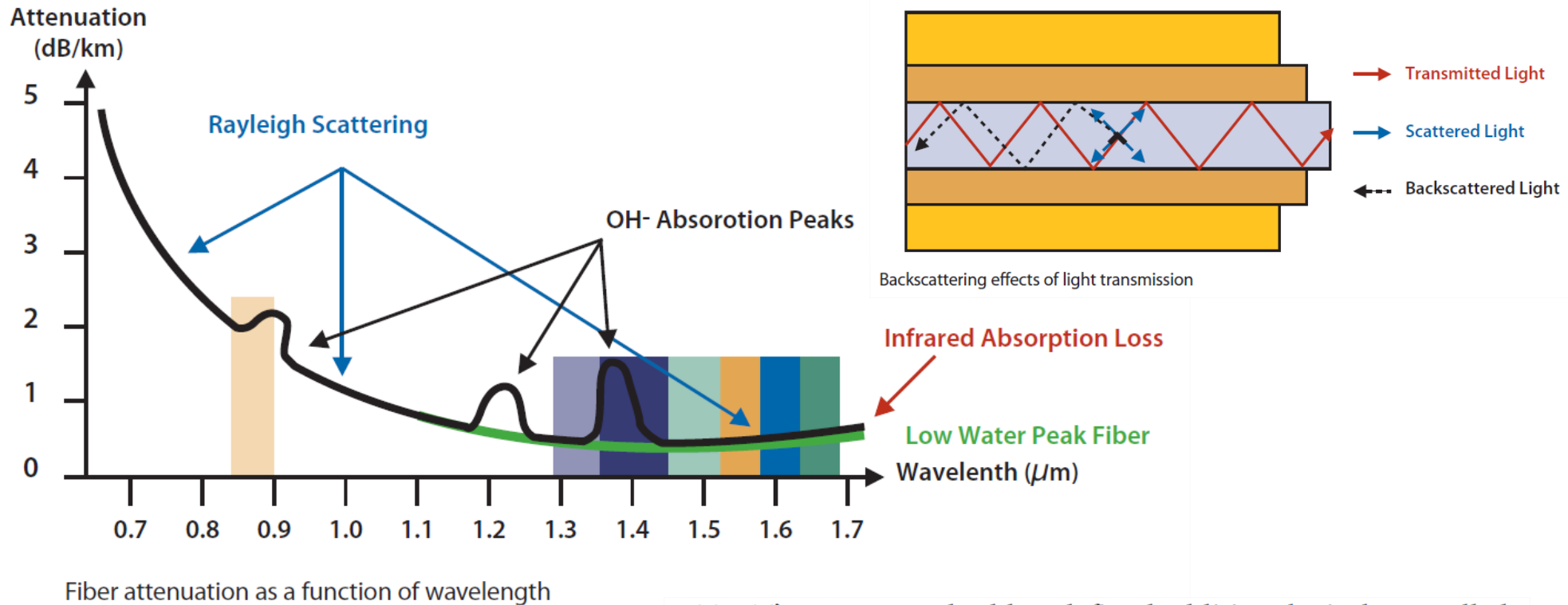
Review of Singlemode and Multimode Fiber

Review of singlemode and multimode fiber

	Multimode	Singlemode
Cost of fiber	Expensive	Less expensive
Transmission equipment	Basic and low cost (LED)	More expensive (laser diode)
Attenuation	High	Low
Transmission wavelengths	850 nm to 1300 nm	1260 nm to 1640 nm
Use	Larger core, easier to handle	Connections more complex
Distances	Local networks (<2 km)	Access/medium/long haul networks (>200 km)
Bandwidth	Limited bandwidth (10 Gb/s over short distances)	Nearly infinite bandwidth (>1 Tb/s for DWDM)
Conclusion	The fiber is more costly, but network deployment is relatively inexpensive.	Provides higher performance, but building the network is expensive.

⇒ **Singlemode fiber (SMF) for long distance at high bit rate**

Fiber optic: a wide spectral transmission window



Conventional band	C-band	1530-1565 nm
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Available bandwidth:

- C-band: 4.4 THz
- L-band: 7.1 THz

ITU-T Recommendation G.Sup39 (12/08)
<http://www.itu.int/rec/T-REC-G.Sup39-200812-I/en>

ITU-T's G.692 standard has defined additional windows, called bands, which are dedicated to DWDM transmission systems.

- | | |
|------------------------------------|------------------------------------|
| ■ 820-880 nm (1st window) | ■ S Band 1460-1530 nm |
| ■ O Band 1260-1360 nm (2nd window) | ■ C Band 1530-1565 nm (3rd window) |
| ■ E Band 1360-1460 nm | ■ L Band 1565-1625 nm |
| | ■ U Band 1625-1675 nm |

Attenuation

As the light signal traverses the fiber, it decreases in power level. The decrease in power level is expressed in dB or as a rate of loss per unit distance (dB/km)

Loss in a system can be expressed as the following:

$$\text{Loss} = \frac{P_{\text{out}}}{P_{\text{in}}}$$

where P_{in} is the input power to the fiber and P_{out} is the power available at the output of the fiber. For convenience, *fiber optic loss* is typically expressed in terms of decibels (dB) and can be calculated using Equation 8-2a.

$$\text{Loss}_{\text{dB}} = 10 \log \frac{P_{\text{out}}}{P_{\text{in}}}$$

Oftentimes, loss in optical fiber is also expressed in terms of decibels per kilometer (dB/km)

Optical power in fiber optic systems is typically expressed in terms of dBm, which is a decibel term that assumes that the input power is 1 mwatt. Optical power here can refer to the power of a laser source or just to the power somewhere in the system. If P in Equation is in milliwatts, Equation gives the power in dBm, referenced to an input of one milliwatt:

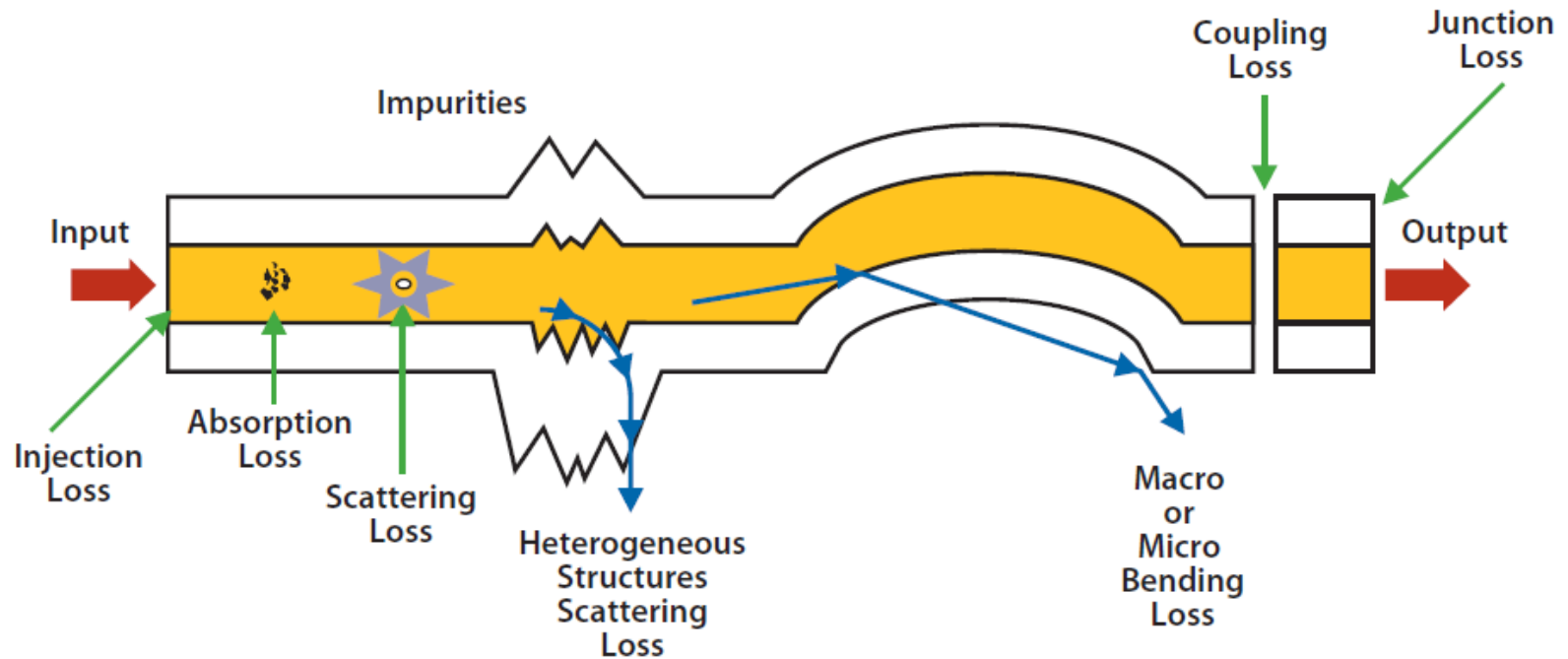
$$P(\text{dBm}) = 10 \log \left(\frac{P}{1 \text{ mW}} \right)$$

With optical power expressed in dBm, output power anywhere in the system can be determined simply by expressing the power input in dBm and subtracting the individual component losses, also expressed in dB. It is important to note that an optical source with a power input of 1 mW can be expressed as 0 dBm. For every 3-dB loss, the power is cut in half.

Consequently, for every 3-dB increase, the optical power is doubled. For example, a 3-dBm optical source has a P of 2 mW, whereas a -6-dBm source has a P of 0.25 mW, as can be verified with Equation.

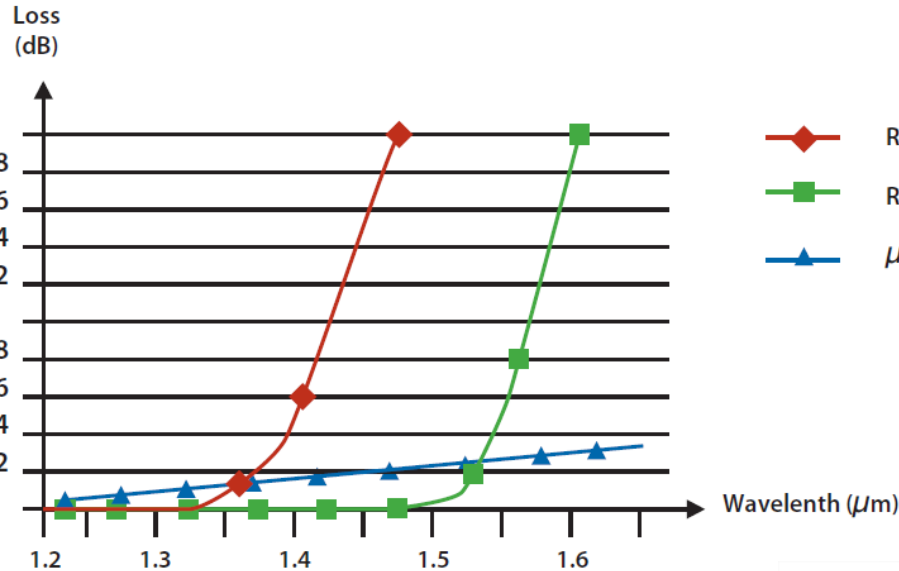
Fiber optic span/link loss mechanism

For a fiber optic span, the effects of passive components and connection losses must be added to the inherent attenuation of the fiber in order to obtain the total signal attenuation. This attenuation (or loss), for a given wavelength, is defined as the ratio between the input power and the output power of the fiber being measured. It is generally expressed in decibels (dB).



Link loss mechanisms

Fiber optic span/link loss mechanism



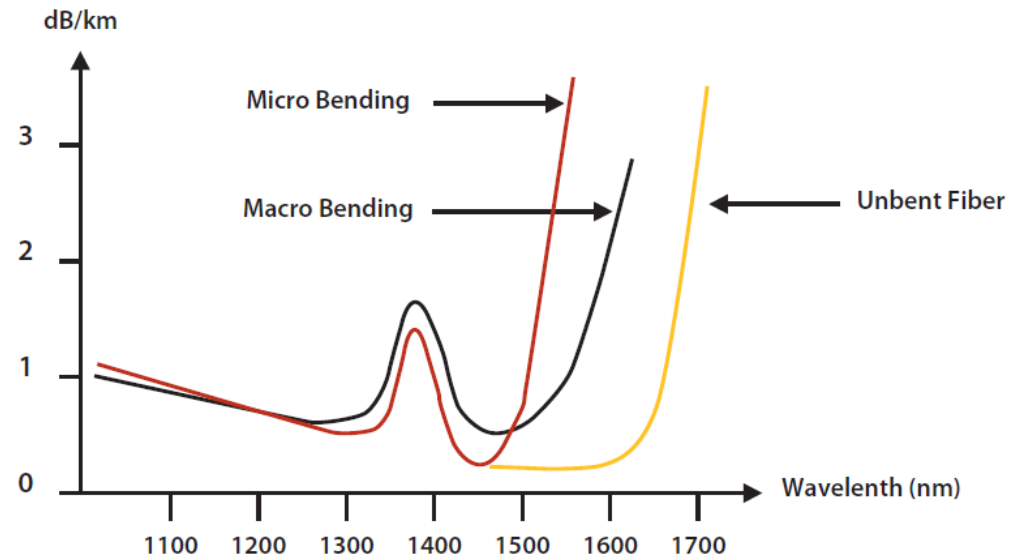
The effects of macro bending on a fiber

Micro Bends and Macro Bends

Micro bends and macro bends are common problems in installed cable systems because they can induce signal power loss.

Micro bending occurs when the fiber core deviates from the axis and can be caused by manufacturing defects, mechanical constraints during the fiber laying process, and environmental variations (temperature, humidity, or pressure) during the fiber's lifetime.

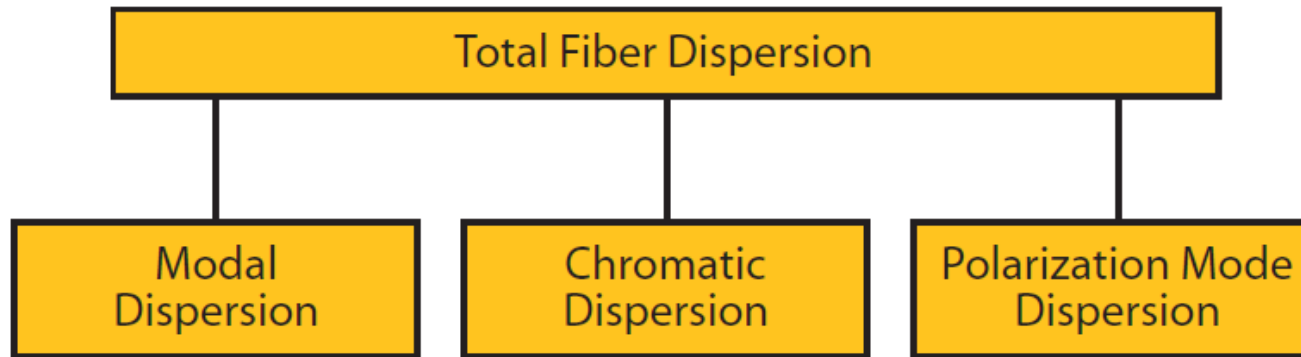
Macro bending refers to a large bend in the fiber (with more than a 2 mm radius). The graph below shows the influence of the bend radius (R) on signal loss as a function of the wavelength. The trace "μc" refers to an ideal fiber with no bending.



Typical attenuation coefficients for bent and unbent fiber

What is dispersion in fiber optic ?

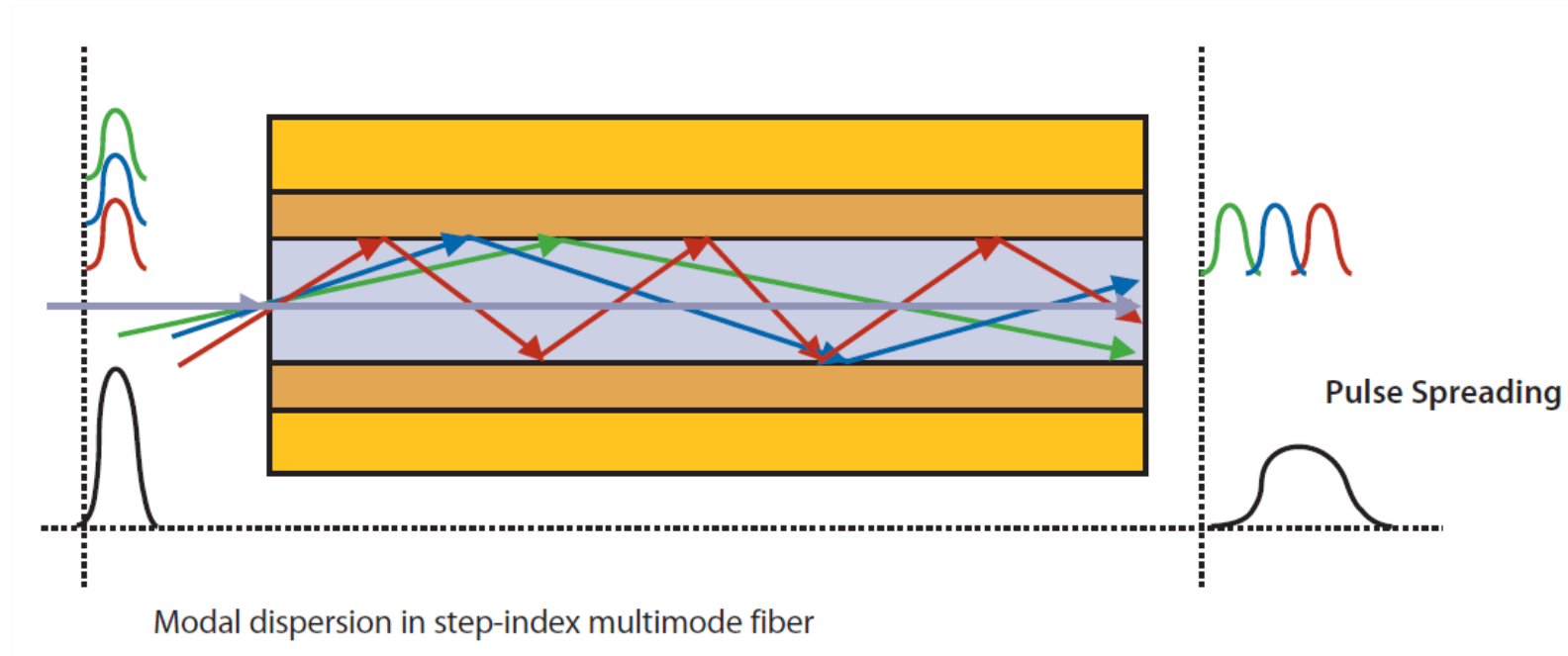
Another factor that affects the signal during transmission is dispersion. Dispersion reduces the effective bandwidth available for transmission. There are three main types of dispersion: modal dispersion, chromatic dispersion, and polarization mode dispersion.



Types of fiber dispersion

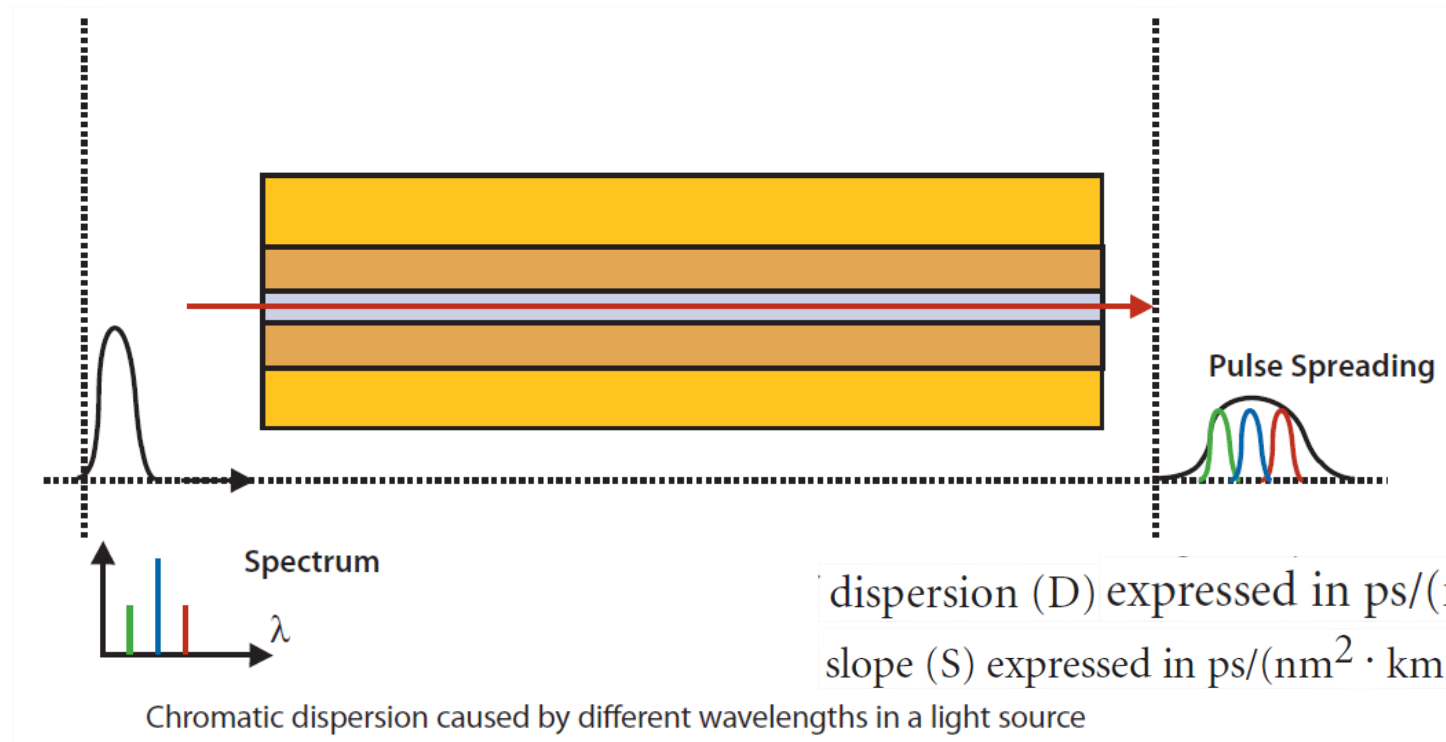
Modal Dispersion

Modal dispersion typically occurs with multimode fiber.



Modal dispersion typically occurs with multimode fiber. When a very short light pulse is injected into the fiber within the numerical aperture, all of the energy does not reach the end of the fiber at the same time. Different modes of oscillation carry the energy down the fiber using paths of differing lengths. For example, multimode fiber with a 50 μm core may have several hundred modes. This pulse spreading by virtue of different light path lengths is called modal dispersion, or more simply, multimode dispersion.

Chromatic Dispersion



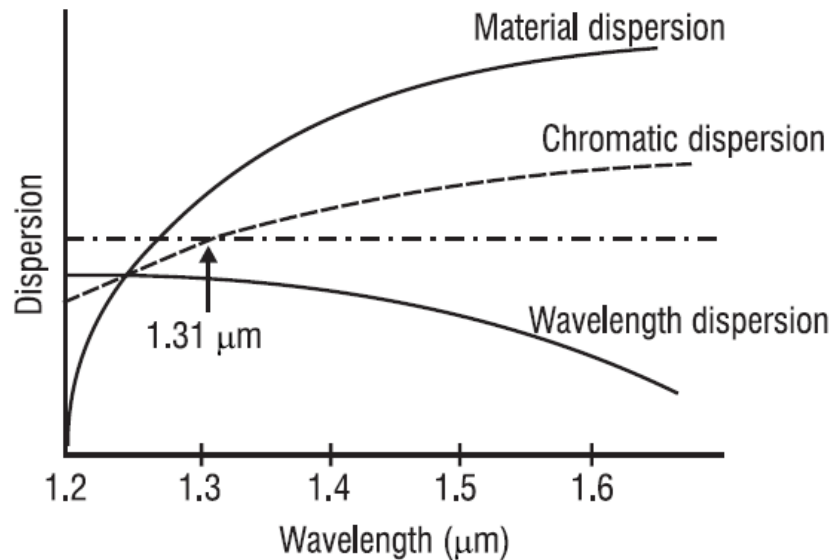
Chromatic dispersion (CD) occurs because a light pulse is made up of different wavelengths, each traveling at different speeds down the fiber. These different propagation speeds broaden the light pulse when it arrives at the receiver, reducing the signal-to-noise ratio and increasing bit errors.

What is dispersion in fiber optic ?

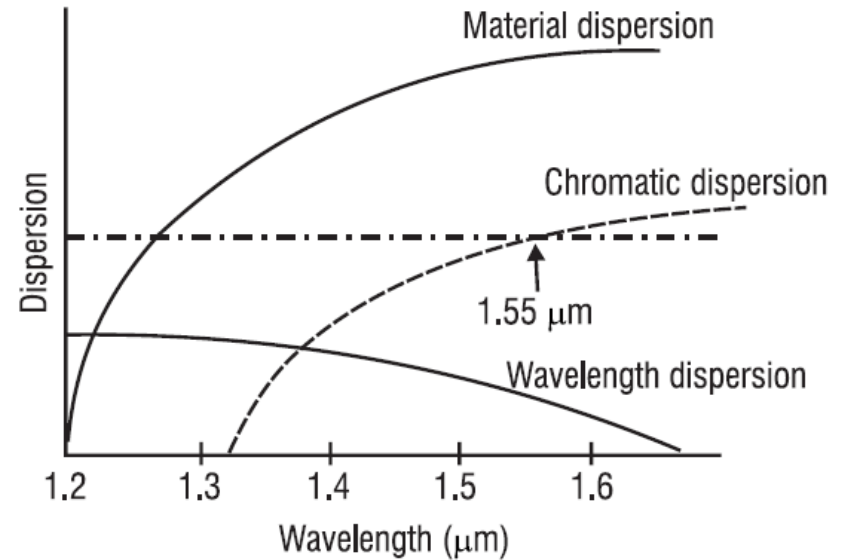
Chromatic Dispersion

$$\Delta t_{\text{chromatic}} = \Delta t_{\text{material}} + \Delta t_{\text{waveguide}}$$

Standard single-mode fiber



Dispersion-shifted fiber



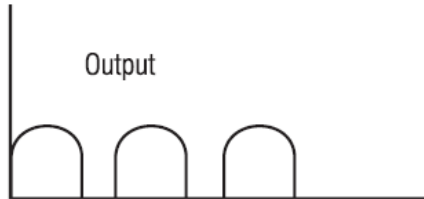
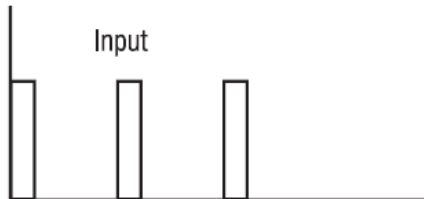
Single-mode versus dispersion-mode versus dispersion-shifted fiber

What is dispersion in fiber optic ?

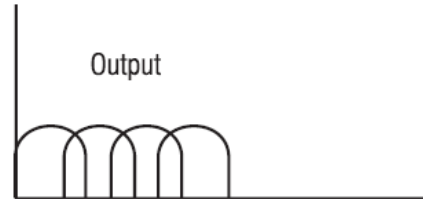
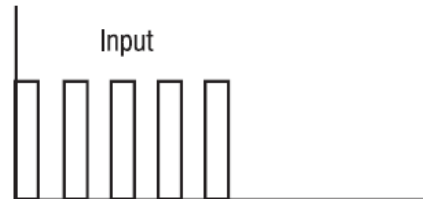
Chromatic Dispersion

$$\Delta t_{\text{total}} = L \times (\text{Dispersion/km})$$

$$\text{BW} = 0.35 / \Delta t_{\text{total}}$$



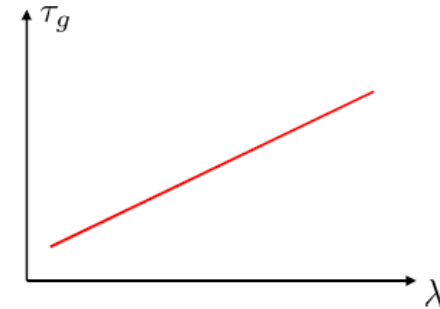
Normal pulses



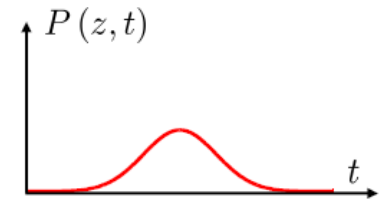
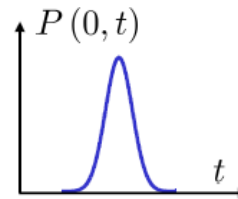
Intersymbol interference

Intersymbol interference

Wavelength
dependence of fibre
effective index



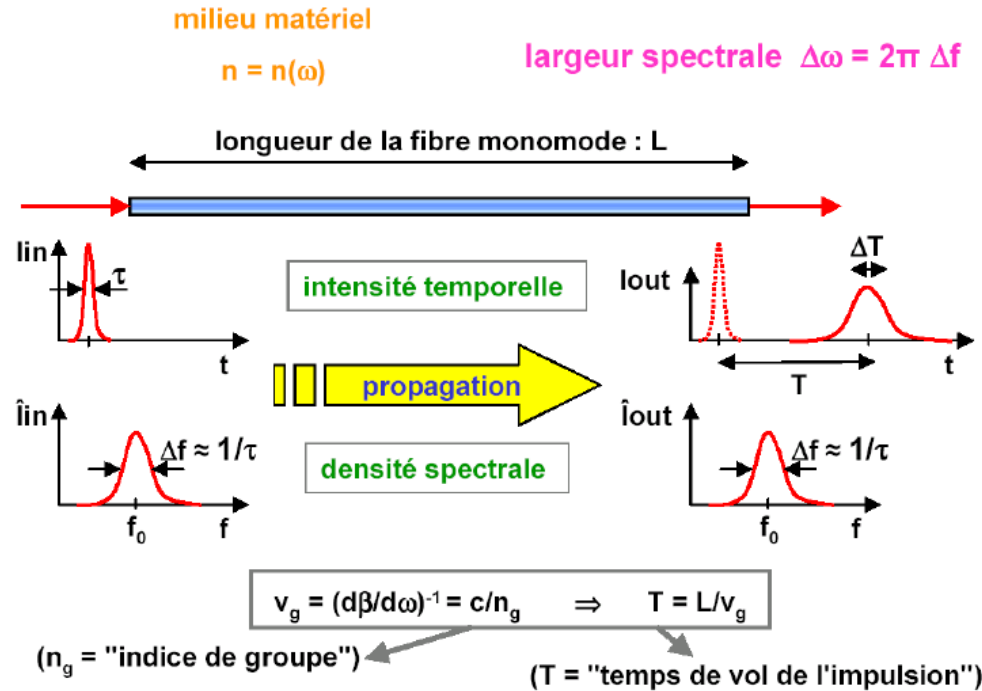
Wavelength dependent
group delay vs.
wavelength



⇒ Dispersion-induced pulse broadening

What is dispersion in fiber optic ?

Chromatic Dispersion



dérivée seconde de β !!

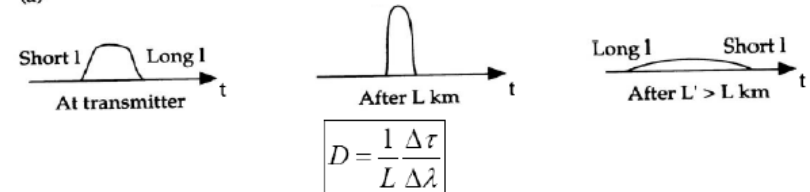
$$\Delta T = \frac{dT}{d\omega} \Delta\omega = \frac{d}{d\omega} \left(\frac{L}{v_g} \right) \Delta\omega = L \frac{d^2\beta}{d\omega^2} \Delta\omega = L \beta_2 \Delta\omega$$

$$\Delta T = \frac{d}{d\lambda} \left(\frac{L}{v_g} \right) \Delta\lambda = L D \Delta\lambda = L D \left(\frac{-\lambda^2}{2\pi c} \right) \Delta\omega$$

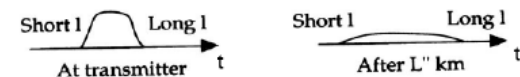
$D = \left(\frac{-2\pi c}{\lambda^2} \right) \beta_2$

D : "paramètre de dispersion" [ps.nm⁻¹.km⁻¹]

(a) $D < 0$: les courtes longueurs d'ondes sont plus lentes



(b) $D > 0$: les grandes longueurs d'ondes sont plus lentes

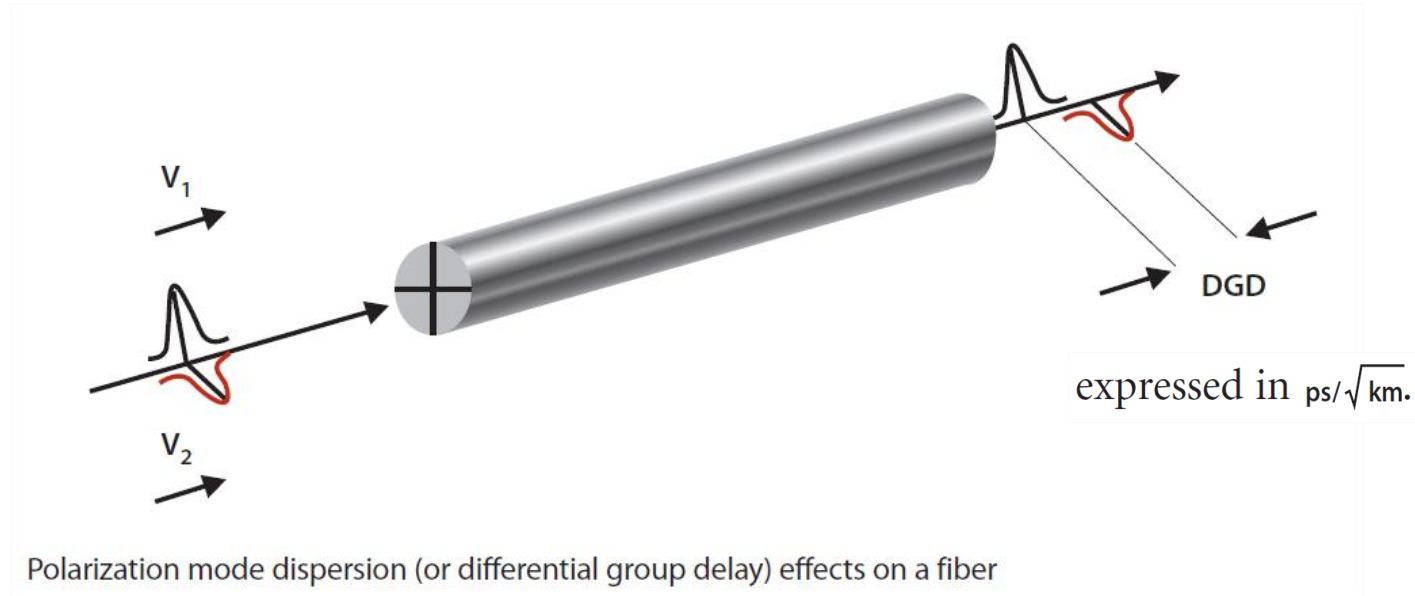


$$\Delta T < T_B = 1/B$$

$$B \Delta T < 1$$

$$B L < 1 / (|D| \Delta \lambda)$$

Polarization Mode Dispersion



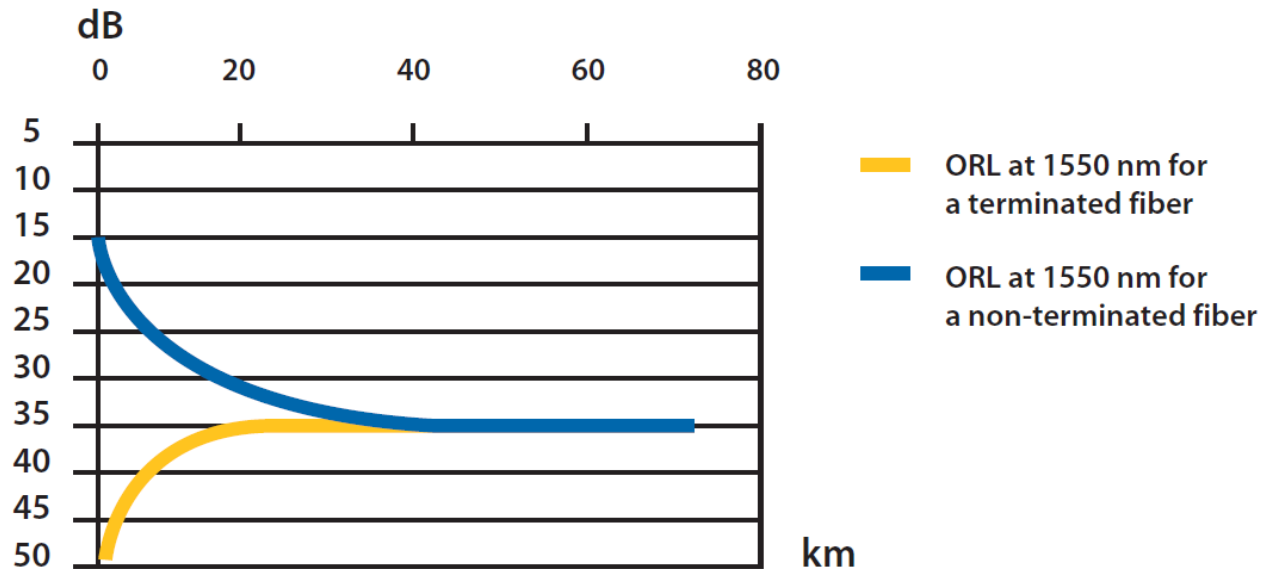
Polarization mode dispersion (PMD) is a basic property of singlemode fiber. It affects the magnitude of the transmission rate. PMD results from the difference in propagation speeds of the energy of a given wavelength, which is split into two polarization axes that are at right angles to each other (as shown in the diagram below). The main causes of PMD are non-circularities of the fiber design and externally applied stresses on the fiber (macro bending, micro bending, twisting, and temperature variations).

What is Optical Return Loss (ORL) in fiber optic ?

ORL is expressed in decibels (dB) and is defined as the logarithmic ratio of the incident power to the reflected power at the fiber origin.

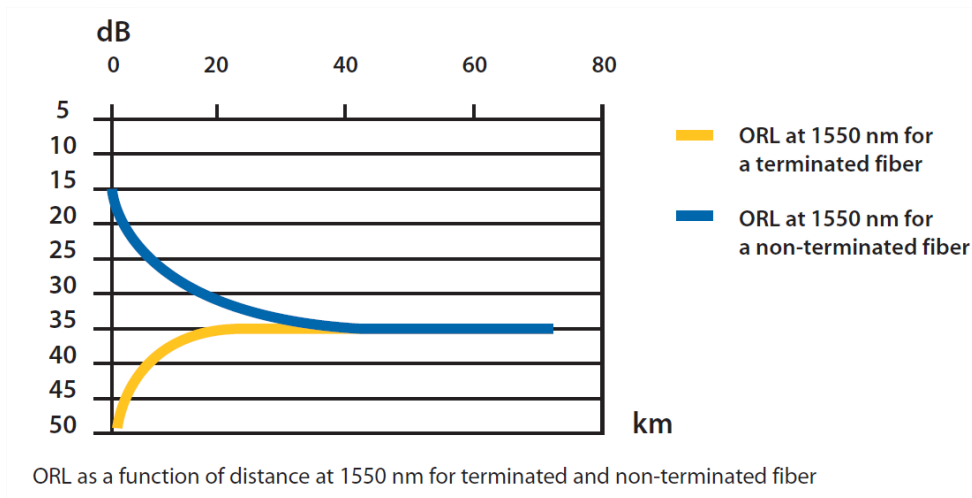
$$ORL = 10 \log \frac{P_e}{P_r} \quad (\geq 0)$$

Where P_e is the emitted power and P_r the reflected power, expressed in Watt (W)



ORL as a function of distance at 1550 nm for terminated and non-terminated fiber

What is Optical Return Loss (ORL) in fiber optic ?



Optical return loss (ORL) represents the total accumulated light power reflected back to the source from the complete optical span. This includes the backscattering light from the fiber itself, as well as the reflected light from all of the joints and terminations.

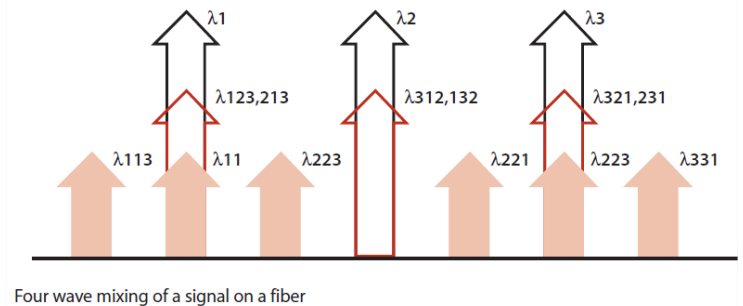
The higher the ORL value, the lower the reflected power; and subsequently, the smaller the effect of the reflection. Therefore, an ORL value of 40 dB is more desirable than an ORL value of 30 dB. It is important to note that ORL is expressed as a positive decibel value whereas the reflectance of a connector is expressed as negative value.

What are Non-Linear Effects in fiber optic ?

Non-linear effects are mainly caused by the high power level and small effective area of the fiber. With an increase in the power level and the number of optical channels, non-linear effects can become problematic factors in transmission systems. These analog effects can be divided into two categories.

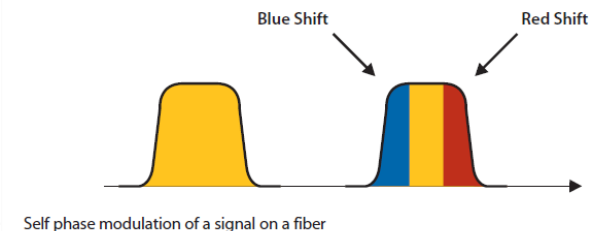
1. Refractive index phenomena: Causes phase modulation by variation in the refractive indexes

- Self phase modulation (SPM)
- Cross phase modulation (XPM)
- Four wave mixing (FWM)



2. Stimulated scattering phenomena: Leads to power loss

- Stimulated Raman scattering (SRS)
- Stimulated Brillouin scattering (SBS)



Summary of transmission effects in fiber optic

Impairment	Causes	Critical Power per Channel	Effects	Compensation
Attenuation	Material absorption/system		– Reduced signal power levels – Increased bit errors	Shorter spans; purer fiber material
Chromatic Dispersion (CD)	Wavelength-dependent group velocity		– Increased bit errors	Use of compensation fiber or modules (DCF/DCM)
Polarization Mode Dispersion (PMD)	Polarization state-dependent differential group delay		– Increased bit errors	New fiber with low PMD values; careful fiber laying; PMD compensators
Four Wave Mixing (FWM)	Signal interference	10 mW	– Power transfer from original signal to other frequencies – Production of sidebands (harmonics) – Channel crosstalk – Increased bit errors	Use of fiber with CD compensators; unequal channel spacing
Self Phase Modulation (SPM) and Cross Phase Modulation (XPM)	Intensity-dependent refractive index	10 mW	– Spectral broadening – Initial pulse compression (in positive CD regimes) – Accelerated pulse broadening (in negative CD regimes) – Channel crosstalk due to "walk-off" effects – Increased bit errors	Use of fiber with CD compensators
Stimulated Raman Scattering (SRS)	Interaction of signal with fiber molecular structure	1 mW	– Decreased peak power – Decreased OSNR – Optical crosstalk (especially in bi-directional WDM systems) – Increased bit errors	Careful power design
Stimulated Brillouin Scattering (SBS)	Interaction of signal with acoustic waves	5 mW	– Signal instability – Decreased peak power – Decreased OSNR – Optical crosstalk (especially in bi-directional WDM systems) – Increased bit errors	Spectral broadening of the light source

The « jungle » of connector types

CONNECTOR AND FIBER TYPES

Single-Fiber Connectors

APC

UPC

	SC Simplex (singlemode/multimode)	
	SC Duplex (singlemode/multimode)	
	FC Simplex (singlemode/multimode)	
	ST Simplex (singlemode/multimode)	
	E2000 Simplex (singlemode/multimode)	
	E2000 Duplex (singlemode/multimode)	
	LX-5 Simplex (singlemode/multimode)	
	LX-5 Duplex (singlemode/multimode)	
	LC Simplex (singlemode/multimode)	
	LC Duplex (singlemode/multimode)	
	MU Simplex (singlemode/multimode)	
	MU Duplex (singlemode/multimode)	

* APC: angled polished connector (8°) – UPC: ultra-polished connector

Multifiber Connectors

	MT-RJ (2 fibers) (singlemode/multimode)	
	MTP/MTA/MPO (4, 8, 12, 24, 36 or 72 fibers) (singlemode/multimode)	

	STANDARD						
Connector Name	FC	SC	E2000	D4	SMA	BICONIC	ST
Typical IL (not APC)	0.2 dB	0.2 dB	0.2 dB	0.2 dB	0.25 dB	0.3 dB	0.4 dB
Number of Fibers	1	1	1	1	1	1	1
Polishing Style	SPC/UPC/APC	SPC/UPC/APC	SPC/UPC/APC	SPC/UPC	FLAT	FLAT	SPC/UPC
Ferrule Type	2.5 mm	2.5 mm	2.5 mm	2.5 mm	2.5 mm	2.5 mm	2.5 mm

SMALL FORM FACTOR					
LC	SMU	SC-DC	LX.5	OPTI-JACK	ESCON
0.2 dB	0.2 dB	0.2 dB	0.3 dB	0.3 dB	0.3 dB
1 or 2	1 or 2	2	1 or 2	2	2
SPC/UPC/APC	SPC	SPC/UPC/APC	SPC/UPC/APC	PRE-POLISHED	SPC
1.25 mm	1.25 mm	2.5 mm	1.25 mm	2.5 mm	2.5 mm

Why cleaning?

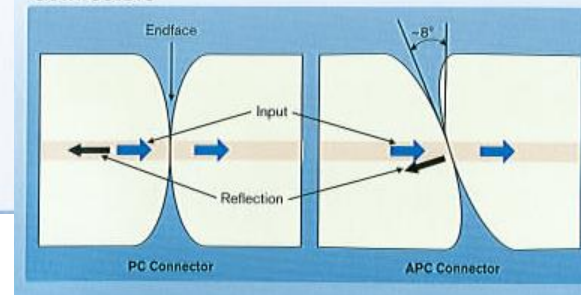
Dirty connector.

This connector was cleaned using alcohol that was not well dried.

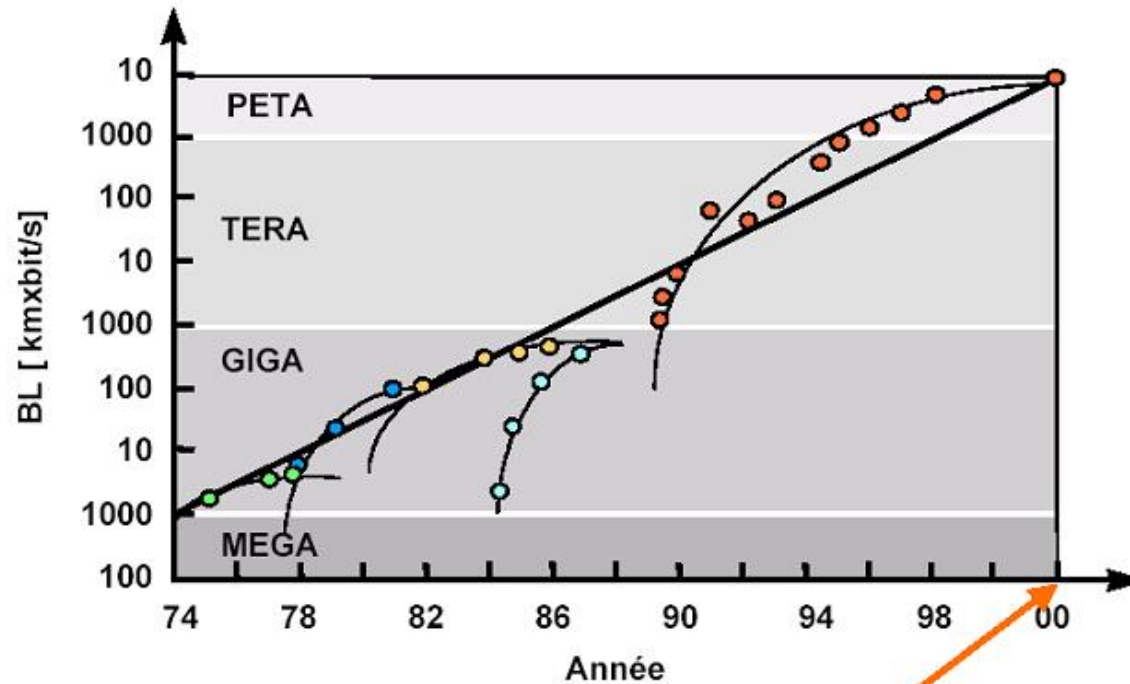
Clean connector.

Why angled?

Connectors



Fiber optic based communication systems: several generations



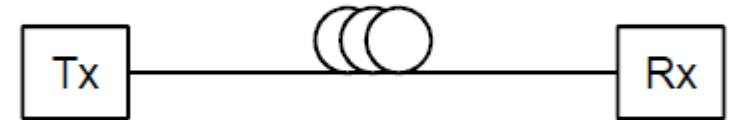
TAT 14/15 - 2 x 3 x 16 x 10 Gbit/s

- 0.8 μm MMF (fibre multimode)
- 1.3 μm SMF (fibre monomode standard)
- 1.5 μm DSF (fibre monomode à dispersion décallée)
- 1.5 μm SMF Coherent detection (fibre monomode standard)
- 1.5 μm SMF DM + WDM + EDFA
(fibres monomodes gérées en dispersion avec amplification optique en ligne)

**Overview on the several generations of optical
communication systems / main issues & concepts**

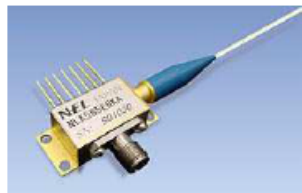
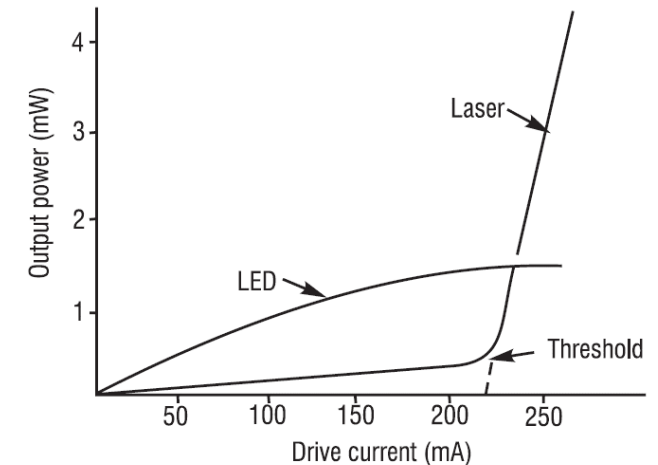
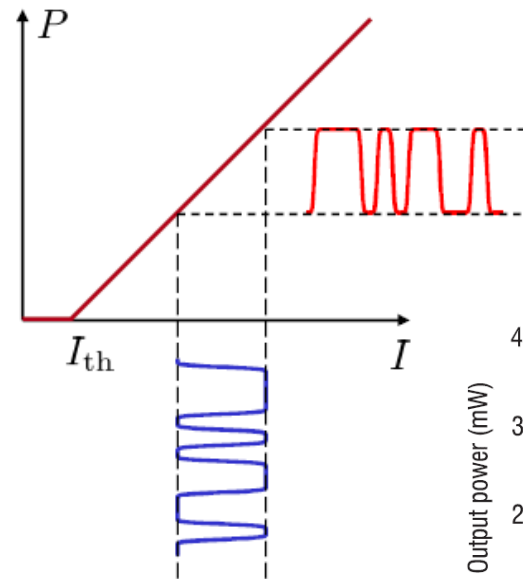
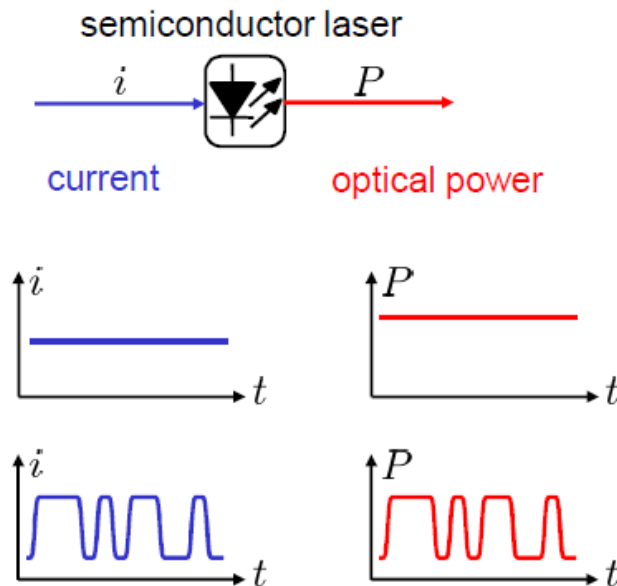
Fiber optic based communication systems: 1st generation

How do we transfer the electrical data to the optical domain?



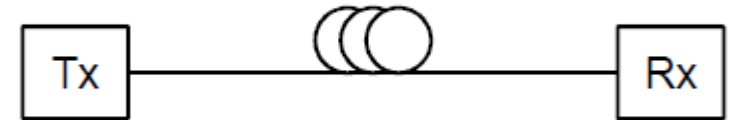
v1.0

Modulation: direct



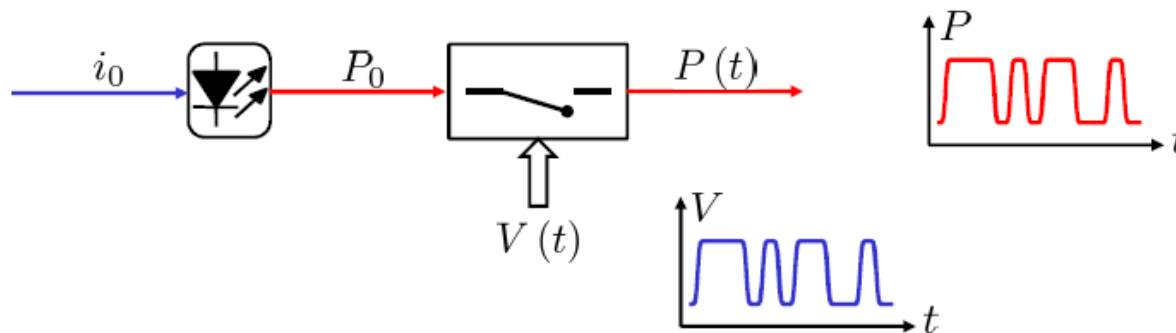
Fiber optic based communication systems: 1st generation

How do we transfer the electrical data to the optical domain?

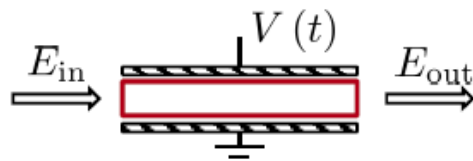


v1.0

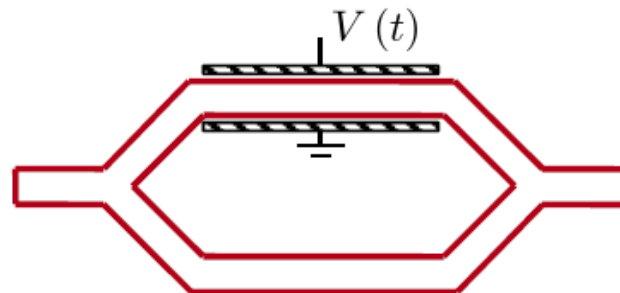
Modulation: external



How?

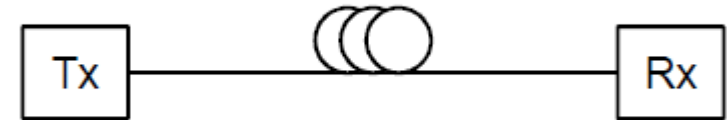


LiNbO₃ crystal



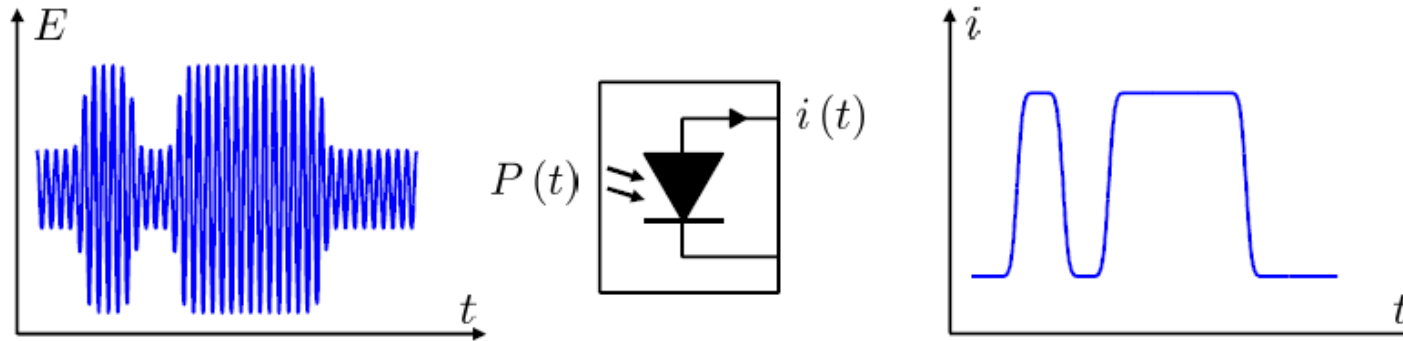
Fiber optic based communication systems: 1st generation

How do we transfer the electrical data to the optical domain?



v1.0

Detection



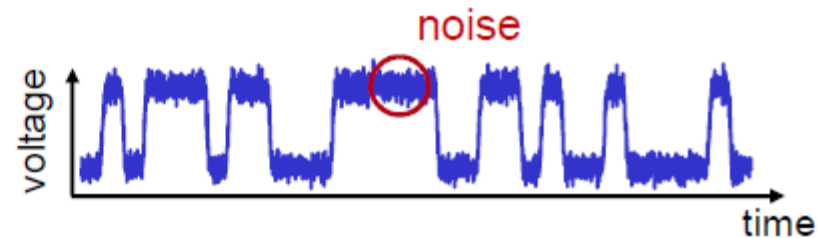
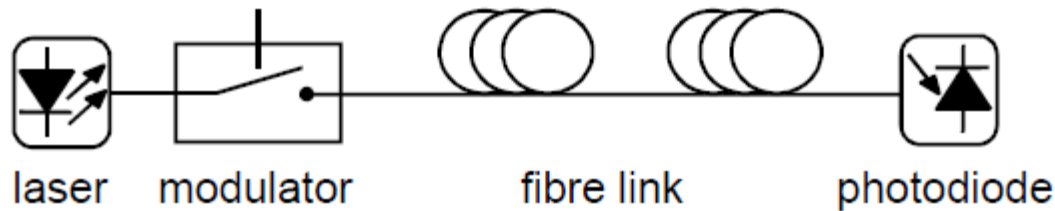
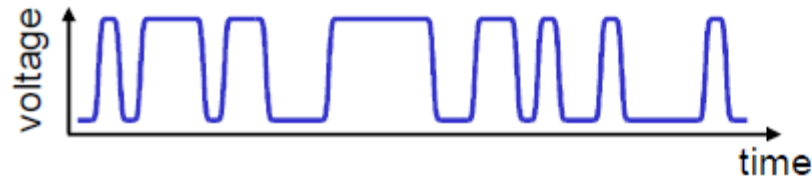
$$i(t) = RP(t)$$

photocurrent

incident optical power

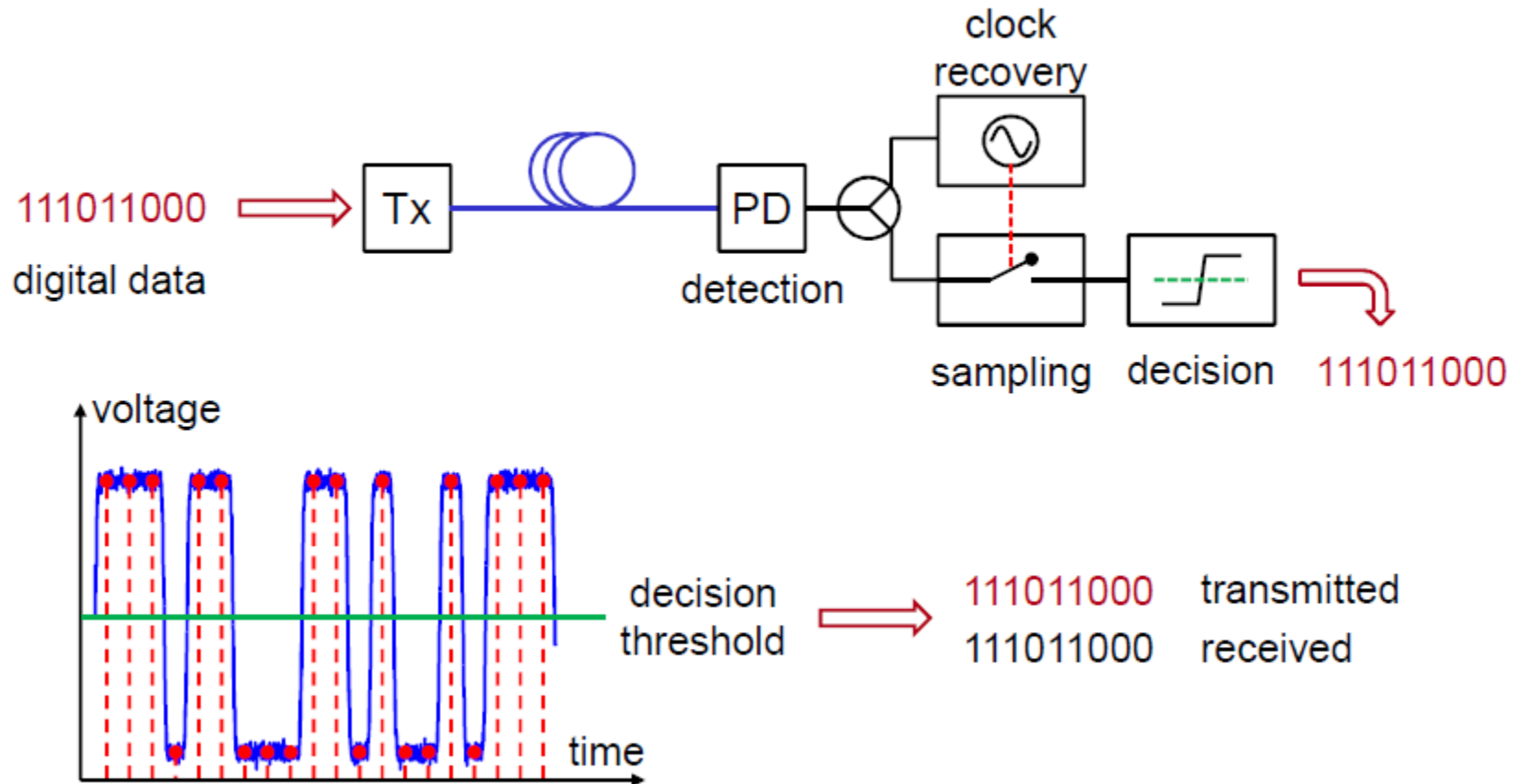
R photodiode responsivity (in A/W)

Fiber optic based communication systems: 1st generation

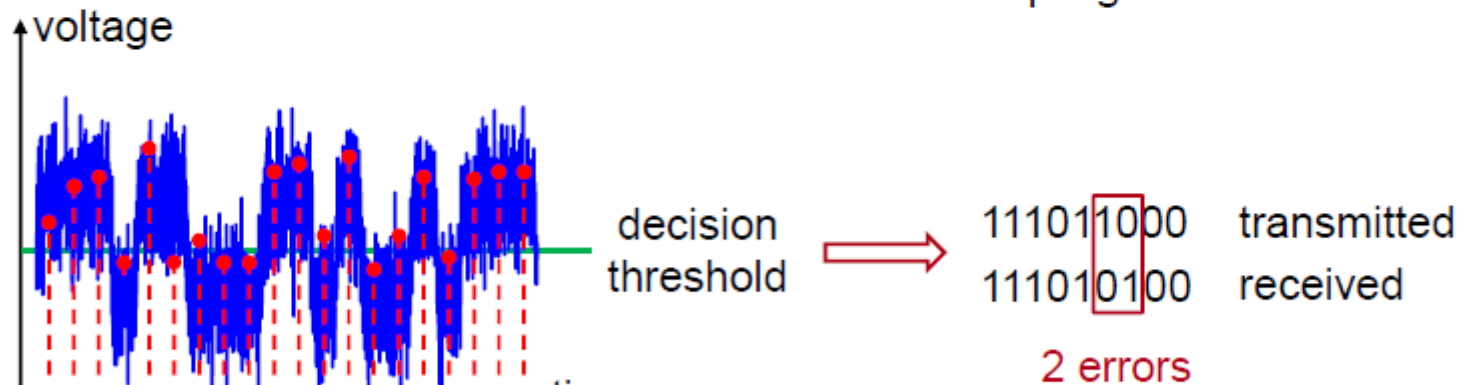
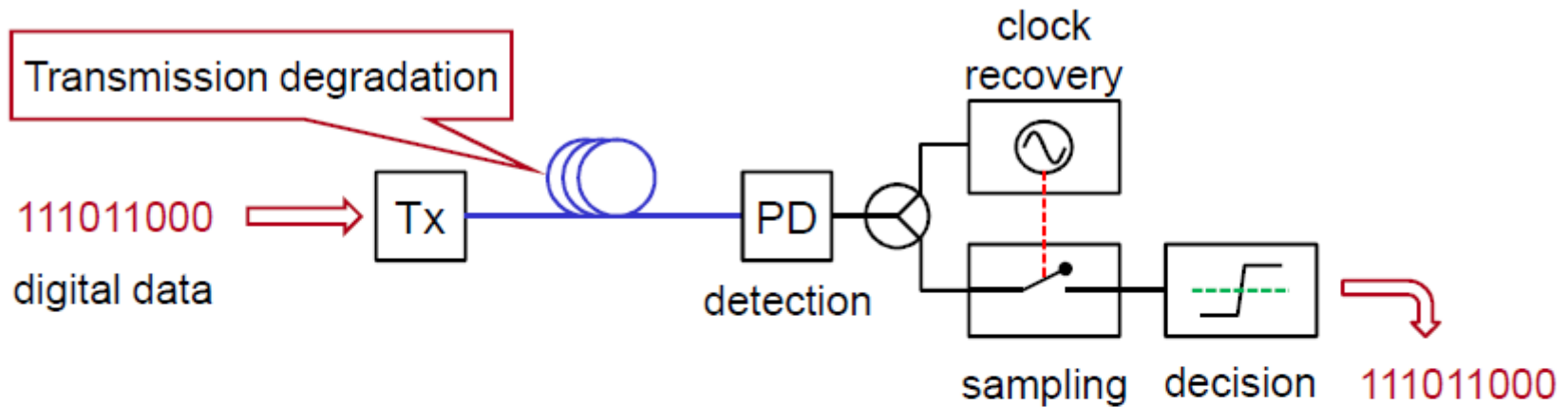


How can noise impact the bit rate of the system?

Digital transmission



Bit-error-ratio

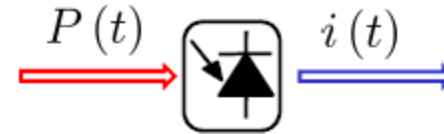


$$\text{BER} = \frac{\text{number of decision errors}}{\text{total number of transmitted bits}}$$

Typical BER target: 10^{-12}

Noise in receivers

What is the physical origin?



Random motions of electrons in conductors $\Rightarrow$ thermal noise

$$\sigma_{\text{th}}^2 = \frac{4k_B T}{R_L} \Delta f$$

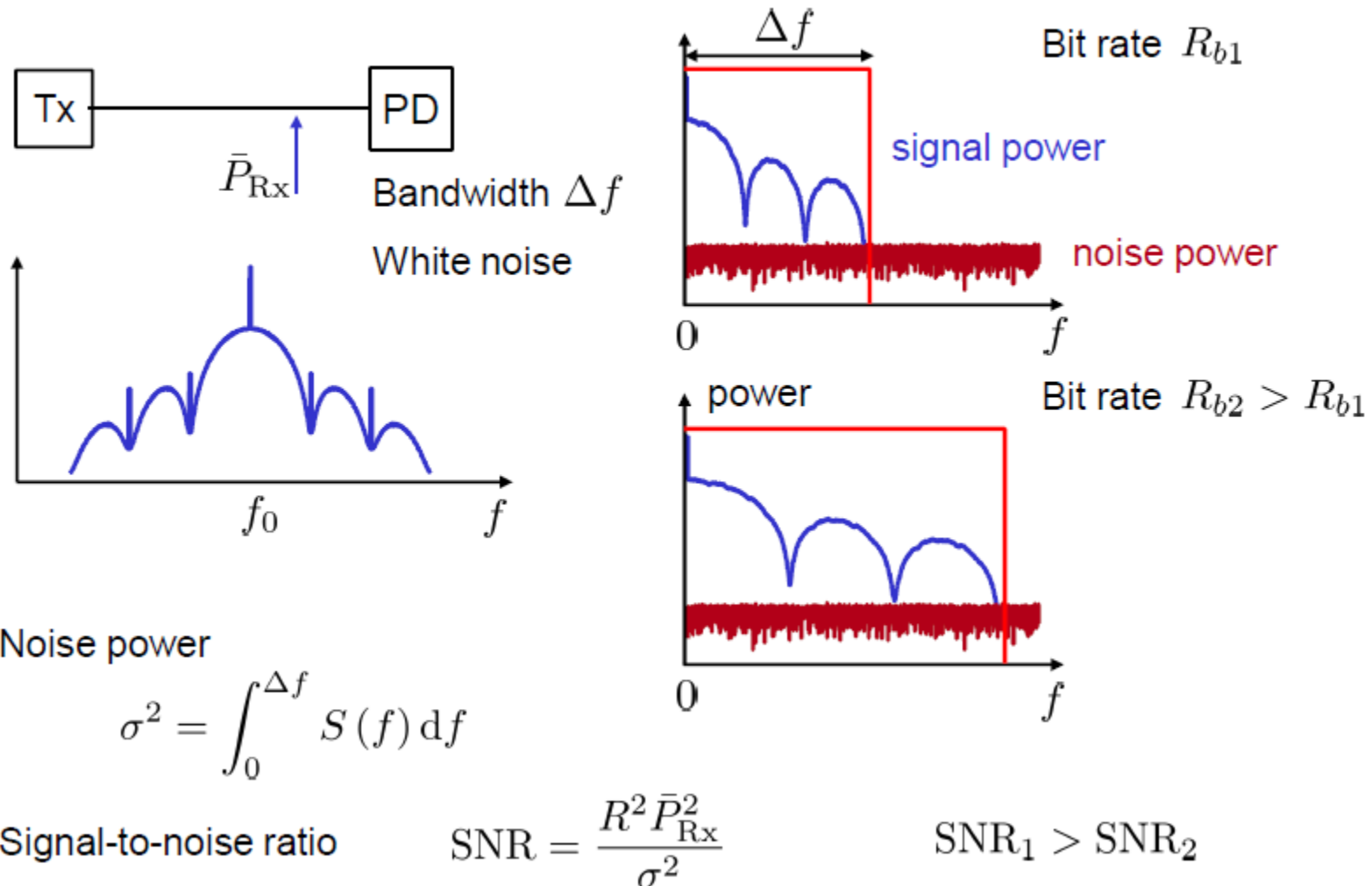
Discrete nature of the photons $\Rightarrow$ shot noise

$$\sigma_s^2 = 2q\bar{I}\Delta f$$

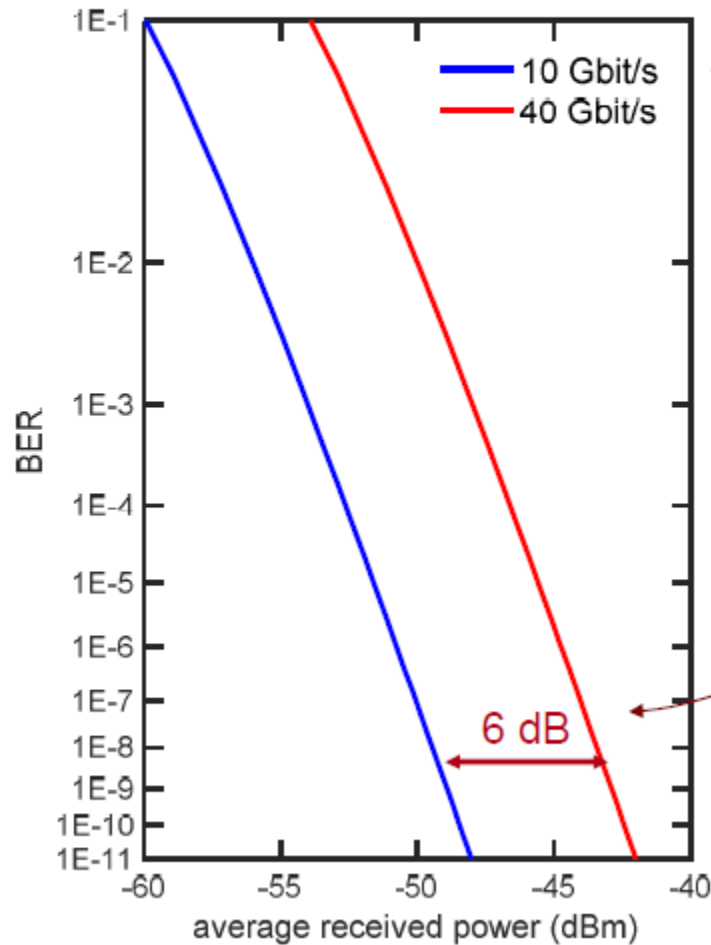
Can it be suppressed?

How does it affects transmission depending on the bit rate?

SNR vs bit rate



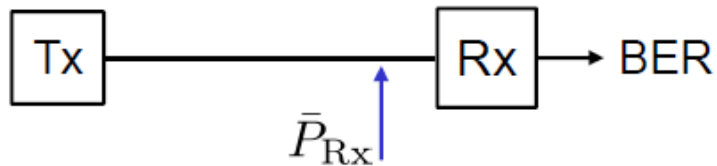
Bit-error-ratio versus received power



bit rate $\times 4$

$\Rightarrow$ The higher the bit rate the higher the required received power for a given BER

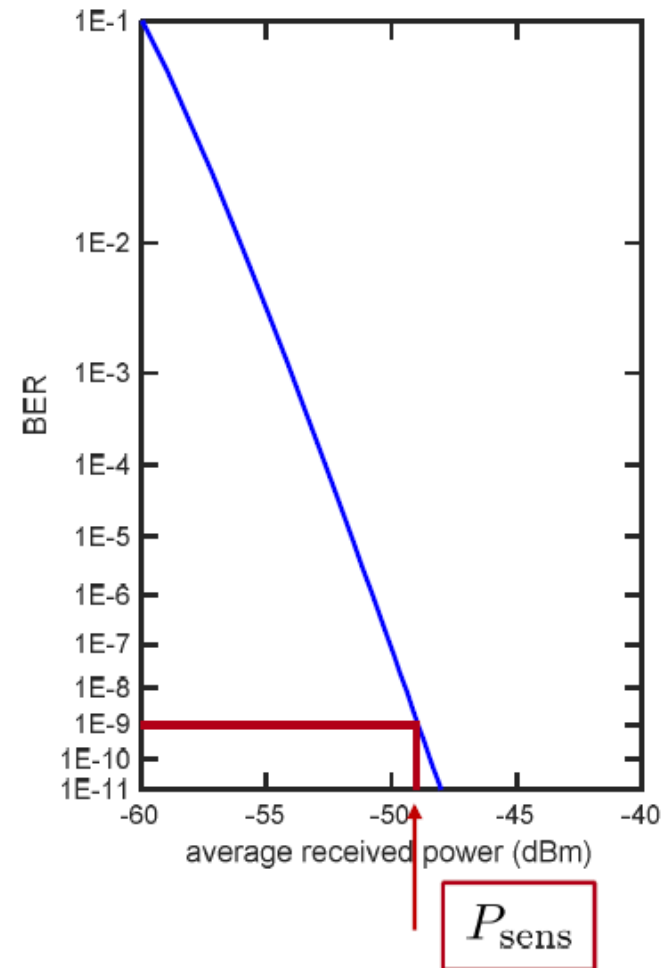
Concept of receiver sensitivity



Receiver sensitivity:

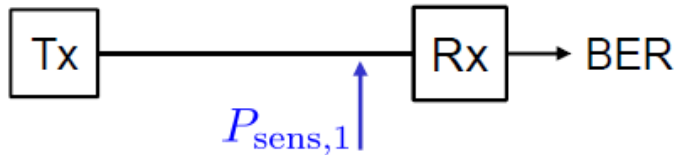
Minimum average power at the input of the receiver in order to reach a specified target BER

Typically BER=10⁻⁹ or 10⁻¹²

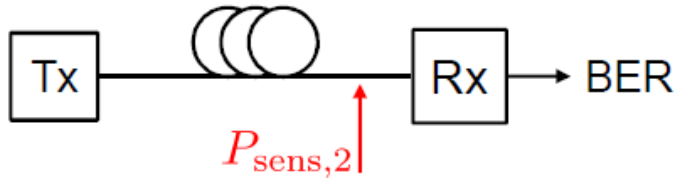


Concept of power penalty

① “Back-to-back” sensitivity



② Introduction of a degradation

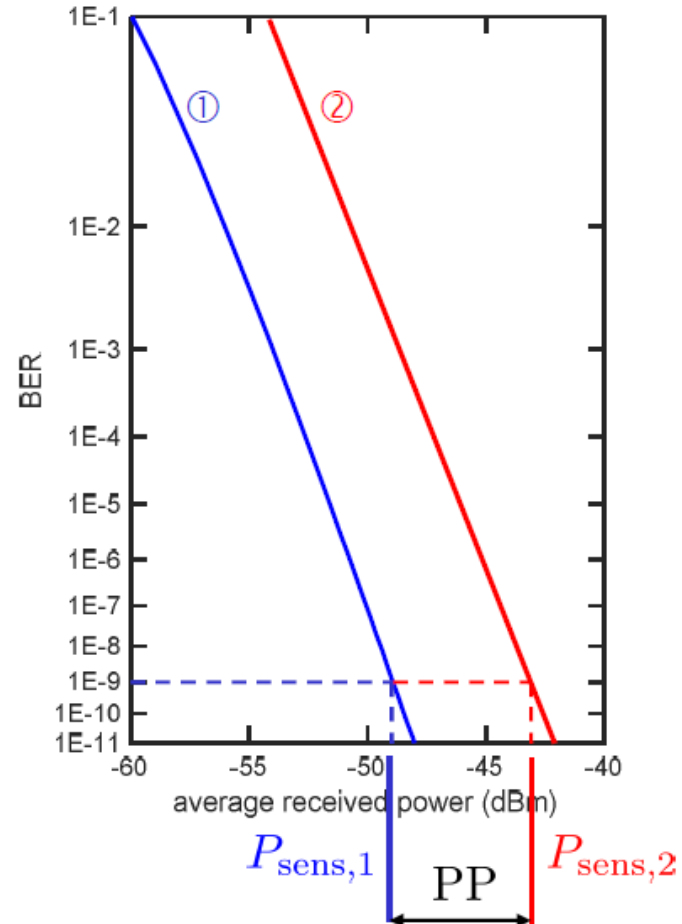


Power penalty: extra power required at the receiver to maintain the target BER

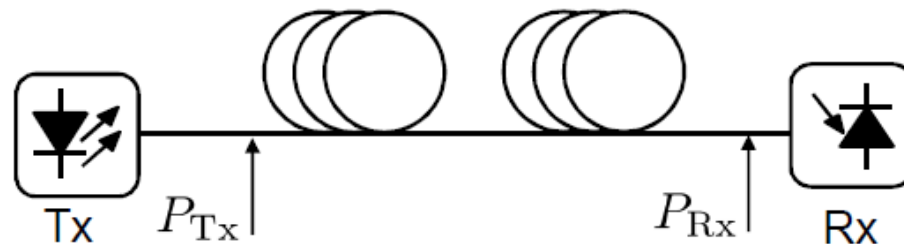
$$PP = P_{sens,2} - P_{sens,1}$$

[dB]

[dB_m]



Reach limit due to attenuation



Maximum power available at the transmitter output

⇒ maximum laser power, loss of components used in the transmitter (modulator)

⇒ e.g. 0 dBm (1 mW)

Absolute minimum power required at the receiver

⇒ receiver sensitivity (minimum receiver power for given BER) + margins

⇒ theoretical minimum corresponds to quantum limit

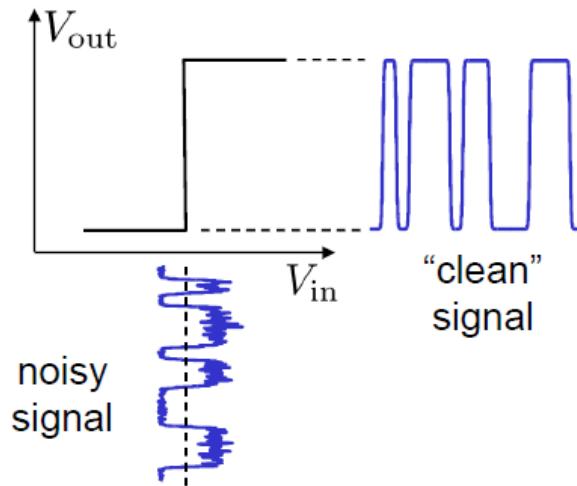
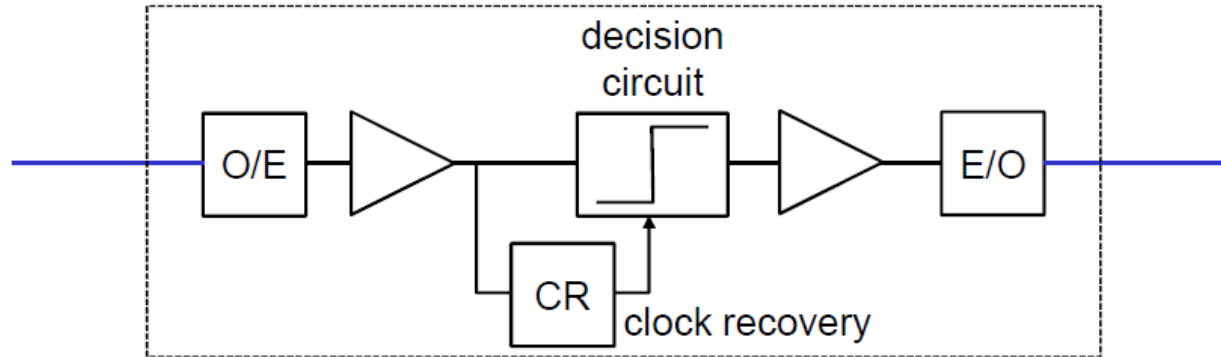
⇒ e.g. -48.9 dBm (13 nW) at 10 Gbit/s for OOK modulation

Maximum tolerable loss:

$$L_{[dB]} = P_{Rx[dB]} - P_{Tx[dB]} \Rightarrow$$

corresponds to 244.5 km of
standard fibre assuming loss of
0.2dB/km

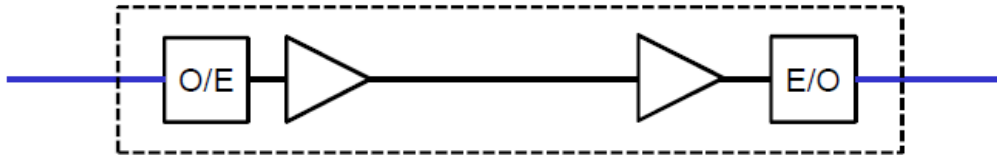
OEO regenerator



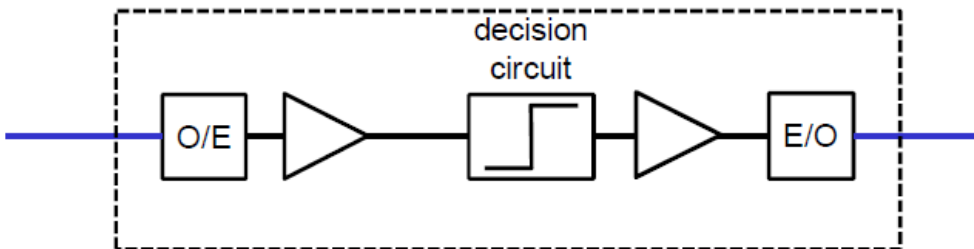
- Reamplification by conversion to the electrical domain
- Can also provide reshaping ("cleaning") of the signal

3R regeneration

1R: re-amplification



2R: re-amplification and re-shaping



3R: re-amplification, re-shaping and re-timing

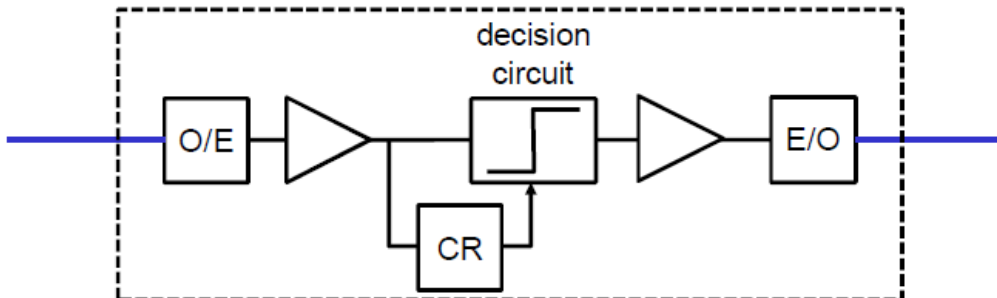
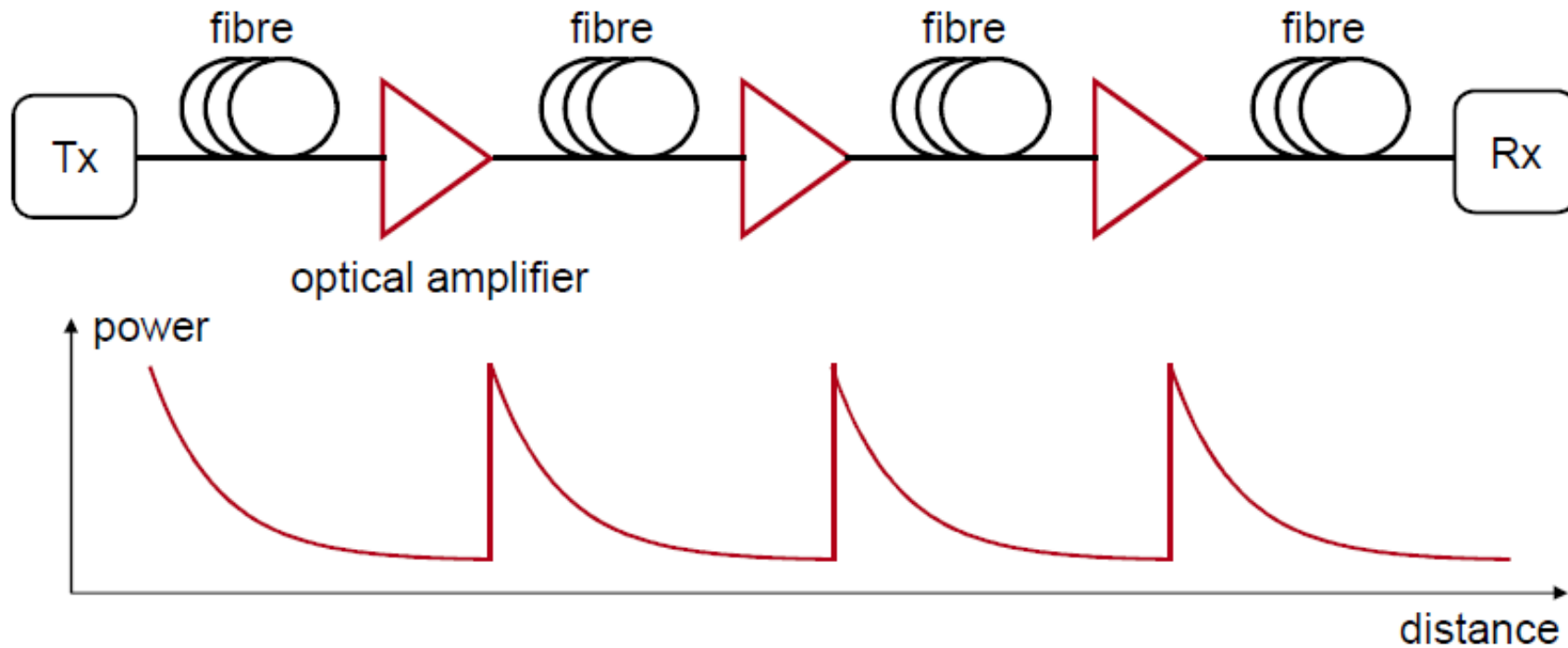


Photo External view of an optical submarine repeater.

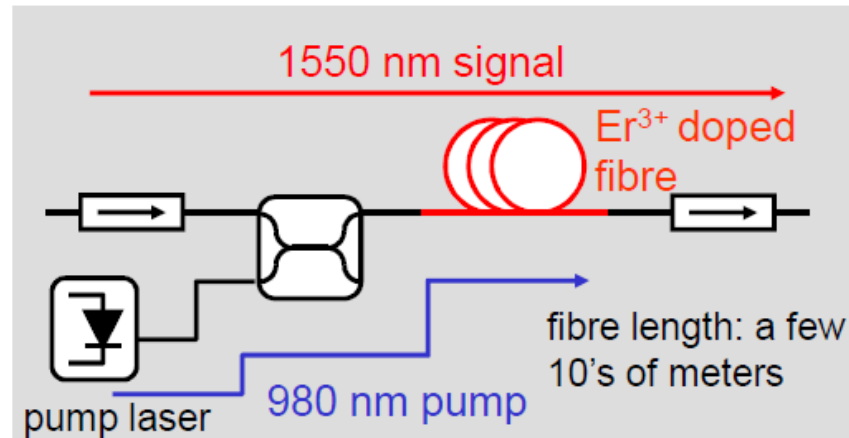
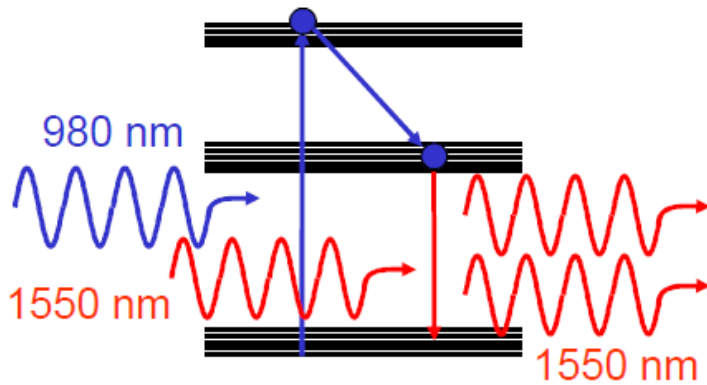
Optically amplified system v2.0



- Typical distance between optical amplifiers: 40-120 km
- Attenuation limitation is overcome using **optical amplification**
- **Long distance systems** are therefore possible

Optical amplification

Erbium (Er^{3+}) doped fibre amplifier (EDFA)

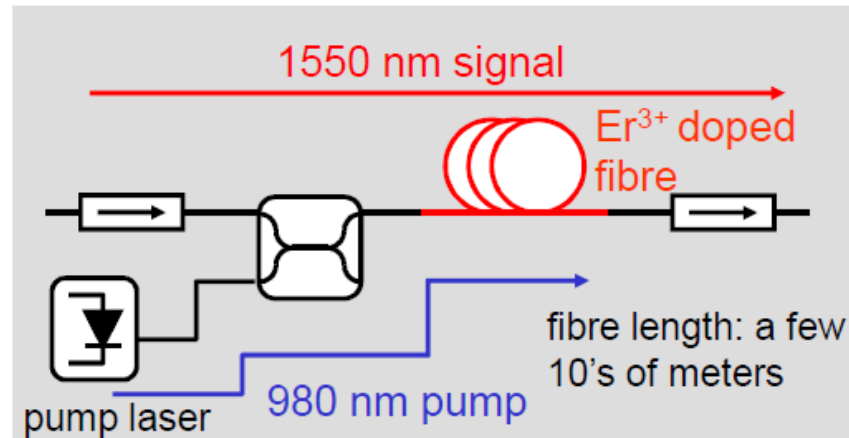
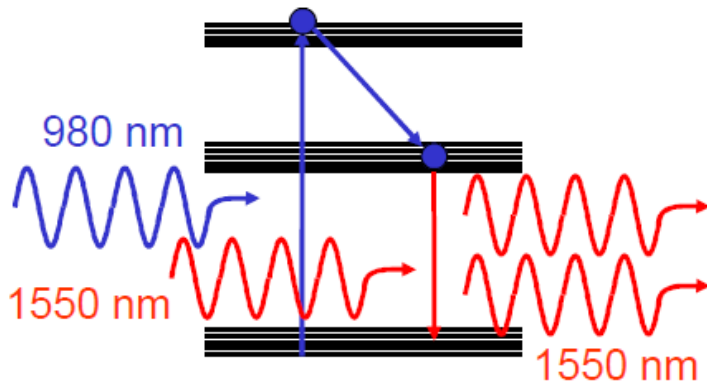


Submarine optical amplifier



Optical amplification

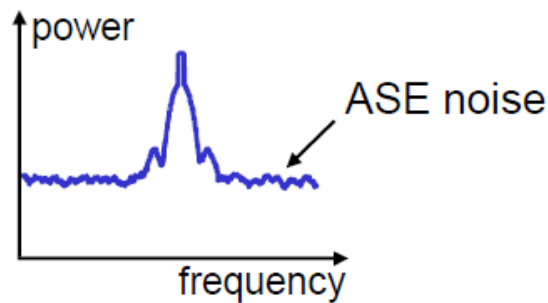
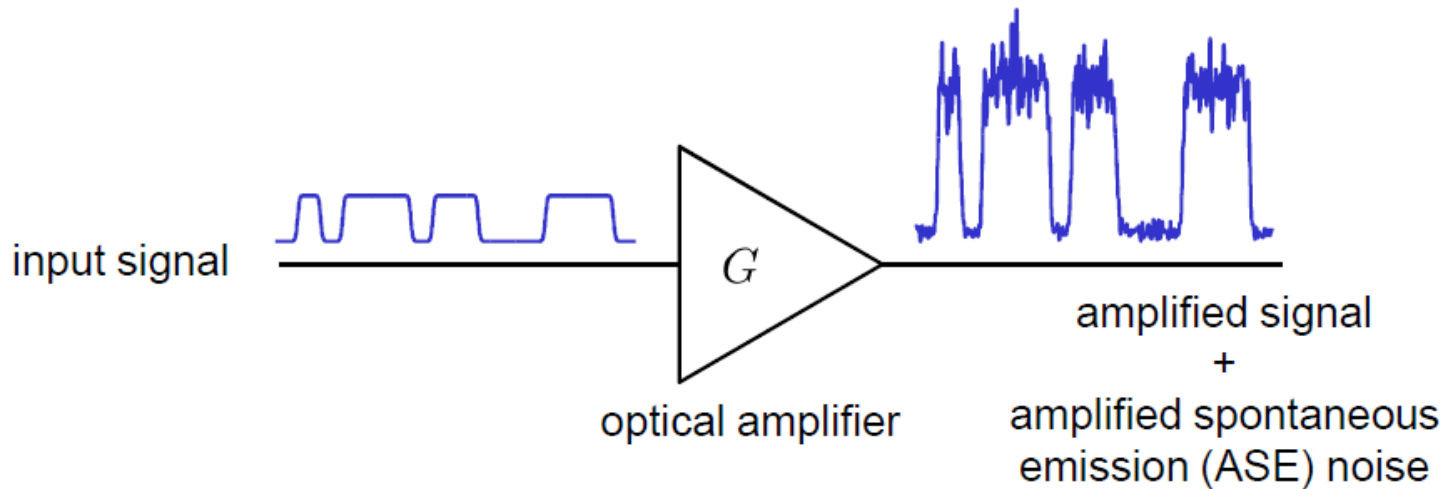
Erbium (Er^{3+}) doped fibre amplifier (EDFA)



Submarine optical amplifier



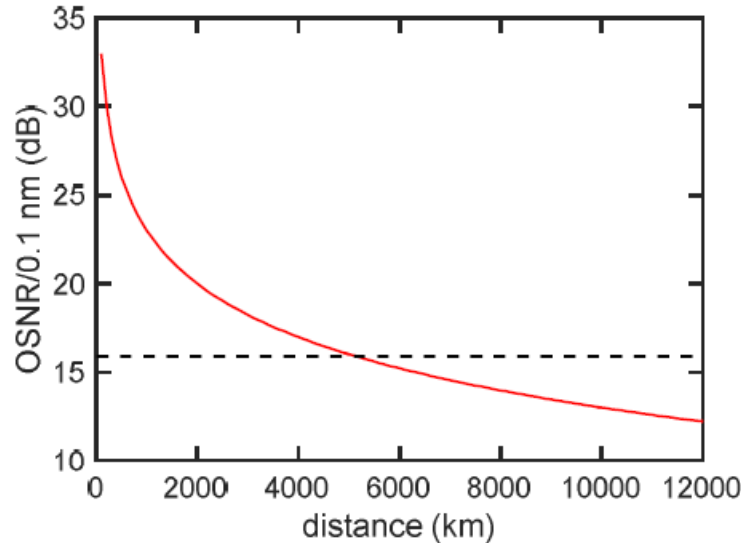
Optical amplifiers add noise



One needs to consider the **optical signal-to-noise ratio**

$$\text{OSNR} = \frac{P_s}{P_{\text{ASE}}}$$

Reach limit due to optical noise

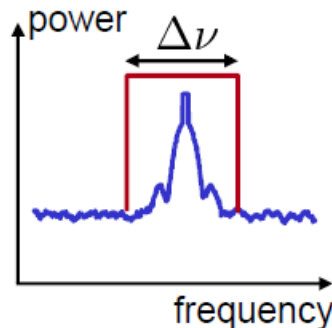


- Adds noise
- Amplifies noise at input



Minimum OSNR at the receiver for a given BER

⇒ Determine the reach of the system



Scaling with bit rate?

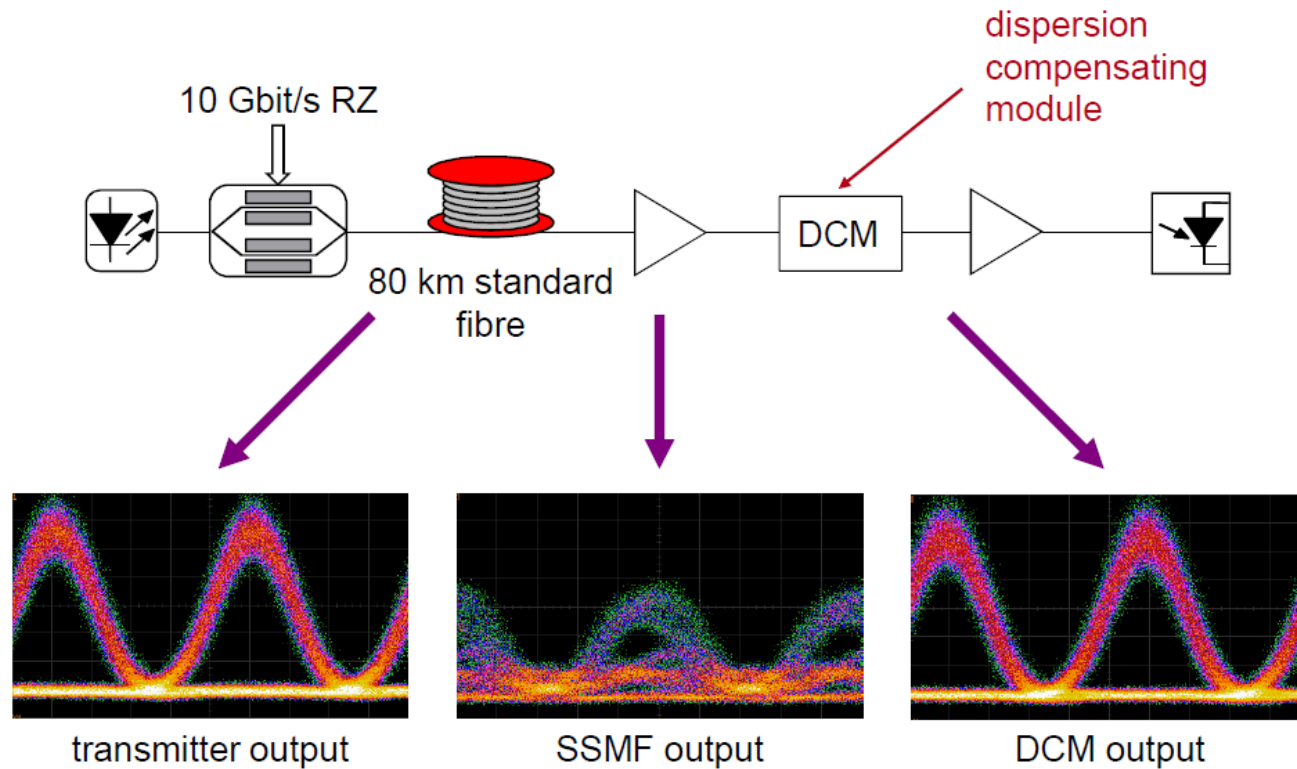
N_0 noise spectral density

$$\text{OSNR} = \frac{P_s}{N_0 \Delta f}$$

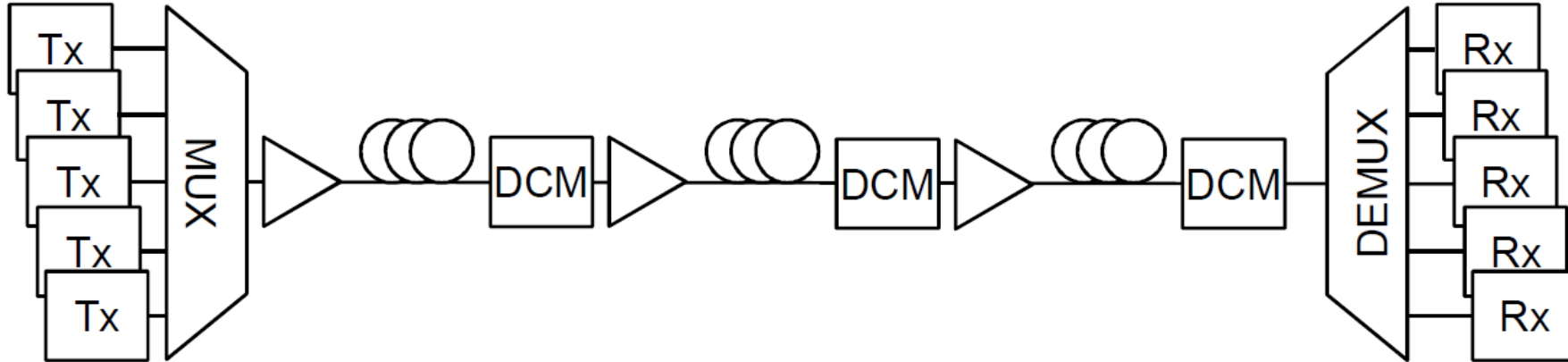
Dispersion compensation



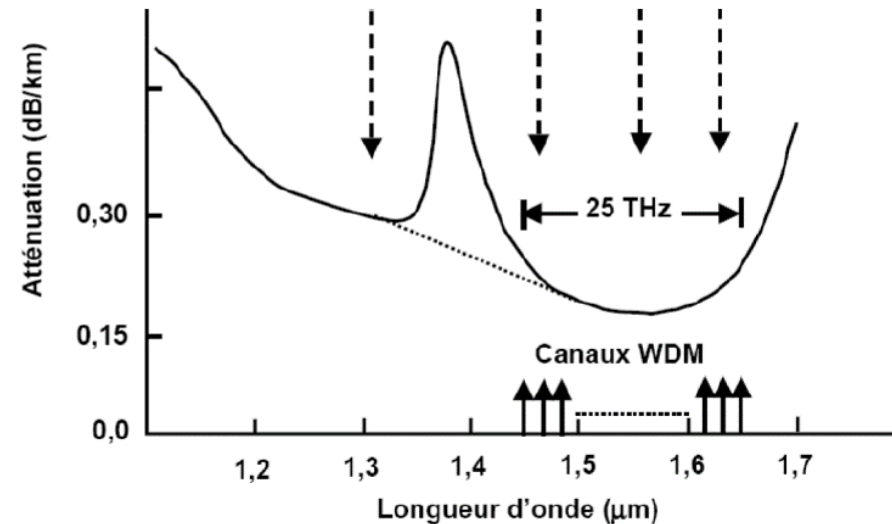
Compensation of dispersion



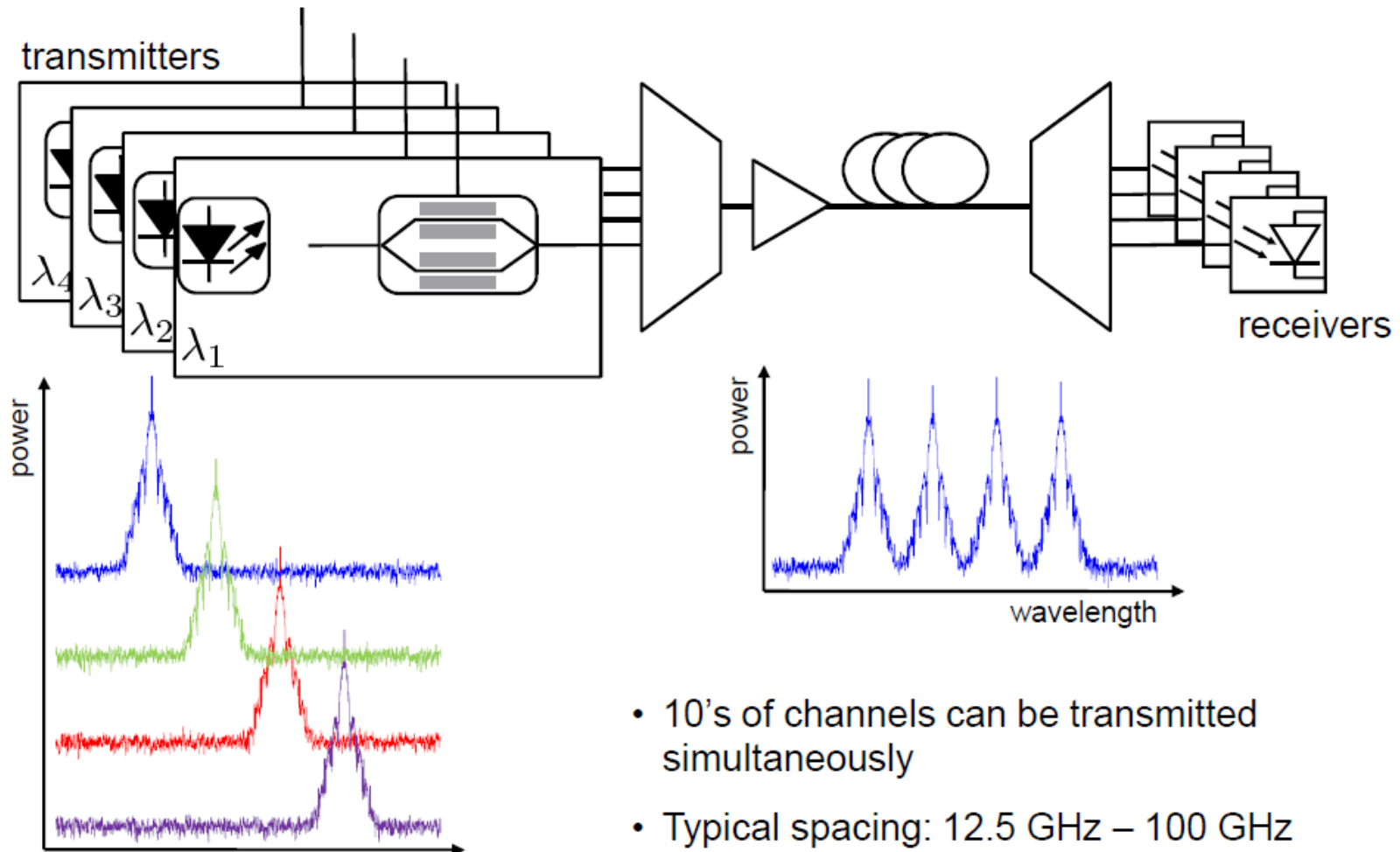
Fiber optic optical communication system: 4th generation



- High bandwidth transmitters and receivers
- Optical amplification
- Dispersion compensation
- Wavelength division multiplexing

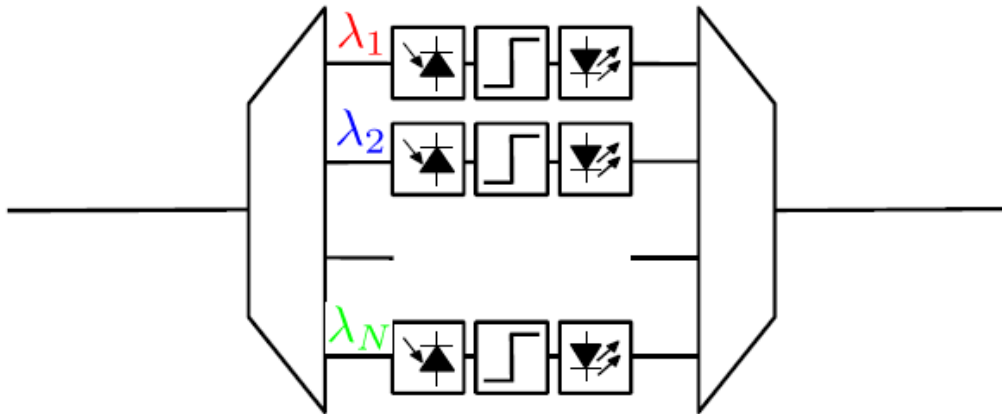


Wavelength multiplexing



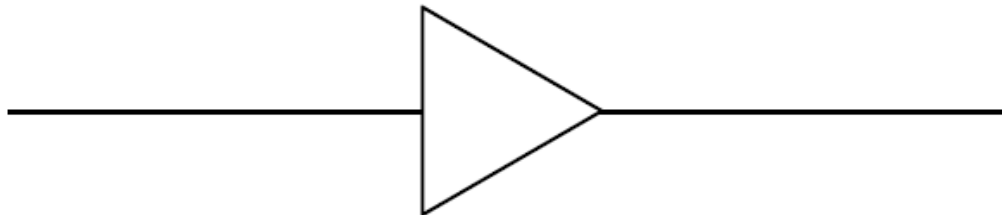
WDM regeneration/amplification

OEO regeneration



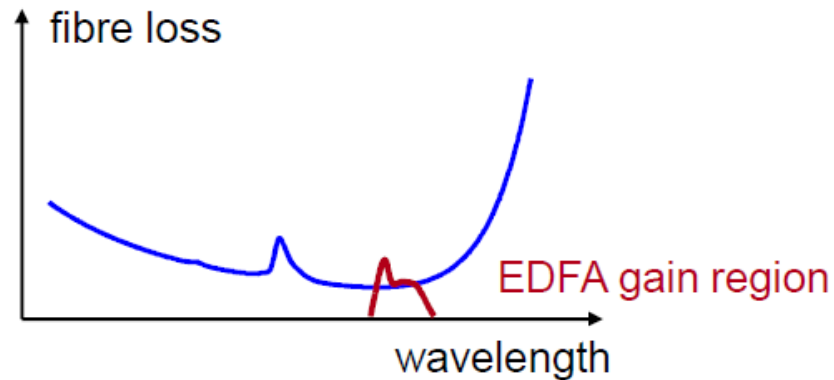
- The different wavelengths need to be de-multiplexed before (up to 3R) regeneration
- As many regenerators as wavelengths needed
- Scalability issue

optical amplification



- One single amplifier can simultaneously amplify all the wavelengths
- No cross-talk between wavelengths in EDFAs
- 2R or 3R functionalities lost

Multi-channel amplification



- EDFA gain region coincides with minimum loss region of optical fibres around 1550 nm
- Several wavelength channels can be amplified simultaneously without interactions in a EDFA

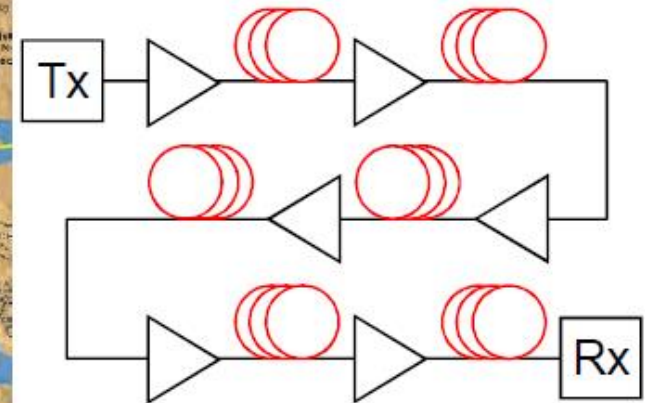
⇒ multi-channel transmission is possible

⇒ wavelength division multiplexing (WDM)

Optically amplified systems



Source: Alcatel Submarine Systems



- Optical amplifiers have revolutionised optical communications
- Used as power booster, in-line amplifier, pre-amplifier
- Lack of optical power is no longer an issue
- However optical amplifiers add noise (noise figure ~5 dB)

Trans-Atlantic system

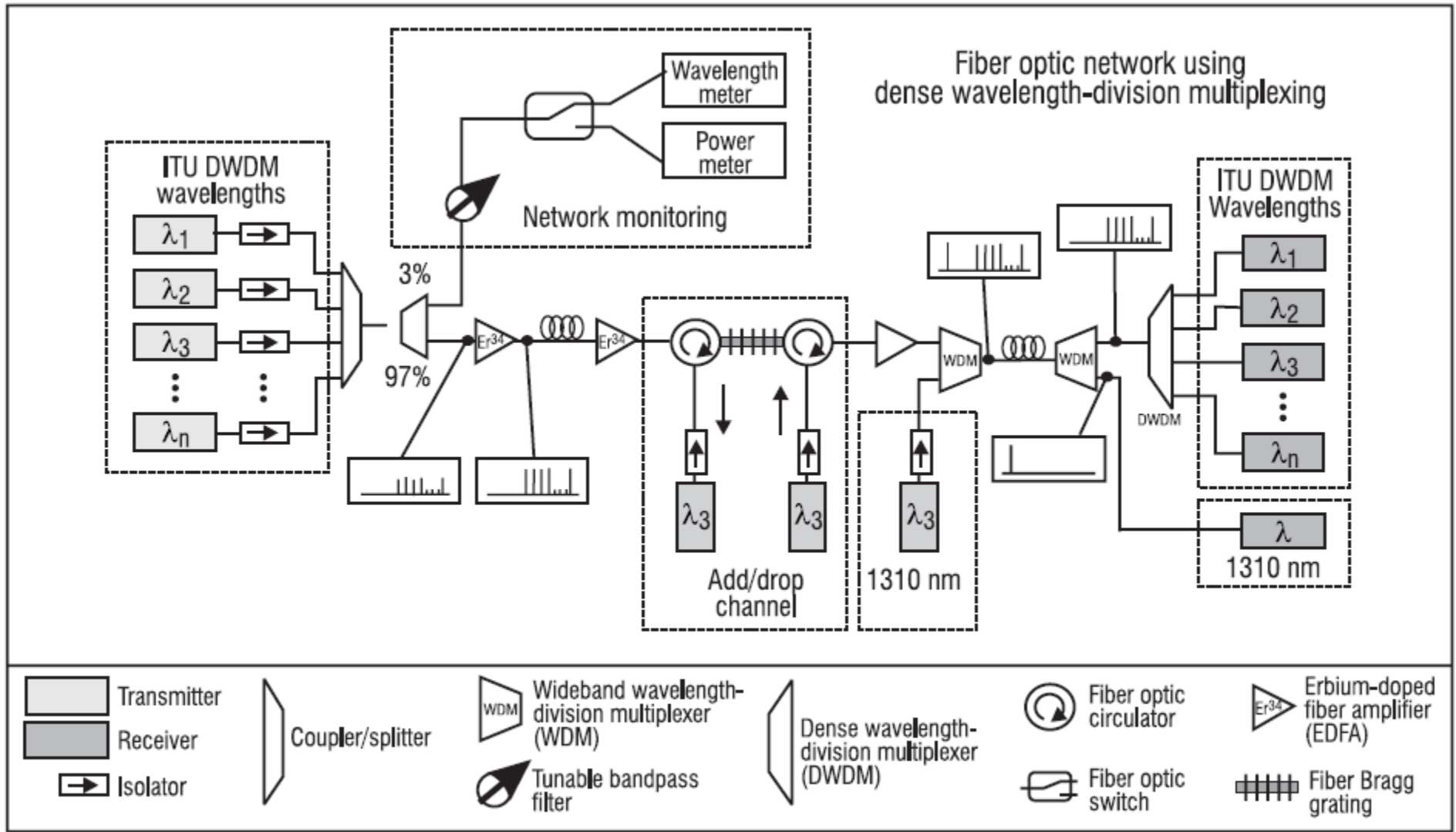
~7500 km

50 km spans

⇒ 150 EDFAs

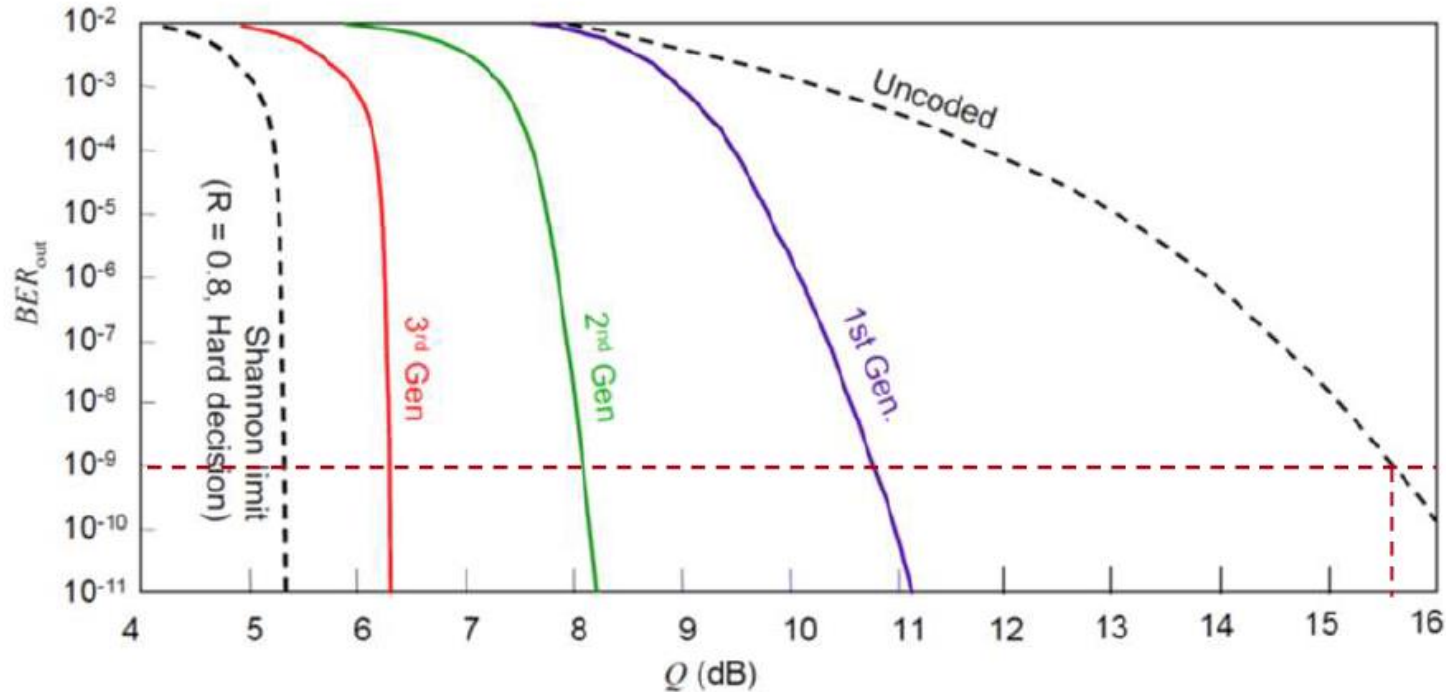
Amplified optical systems are no longer limited by power budget

Fiber optic optical communication system: next generations



Typical DWDM transmission system (Courtesy of Newport Corporation)

Forward error correction

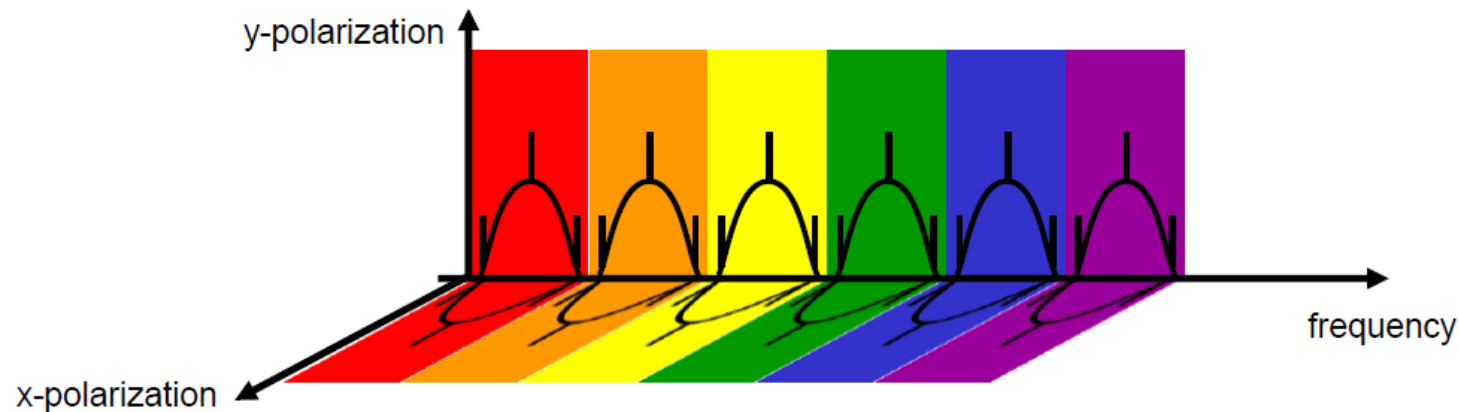


Q : other way to express the BER

$Q=6$ (15.6 dB) for $BER=10^{-9}$

Polarization Division Multiplexing

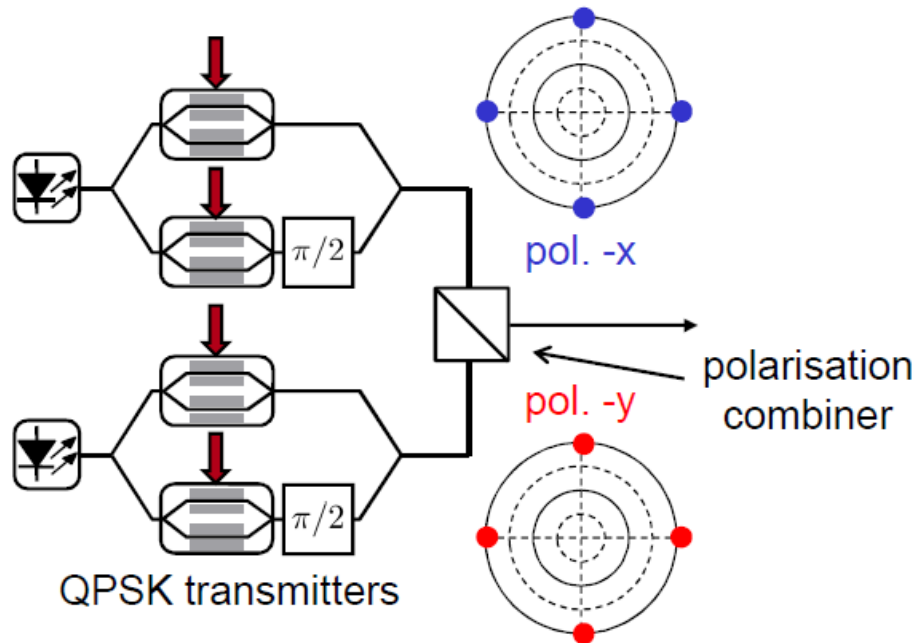
- the optical wave can be split into two orthogonal polarizations
- each polarization can carry an independent signal and can be viewed as a virtual transmission channel within the fiber
- these signals can be separated using polarization diversity receivers



(Source : Agilent technologies 2010)

→ PDM increases the spectral efficiency by a factor of 2

Polarisation-multiplexed QPSK



electrical
OOK signals

4x25 Gbit/s

optical QPSK
signals

2x25 Gbaud
2x50 Gbit/s

Advanced Optical Modulation Formats

Choice of the right modulation format is key to building flexible and cost-effective high-capacity optically routed wavelength division multiplexed fiber networks.

By PETER J. WINZER, Senior Member IEEE, AND RENÉ-JEAN ESSIAMBRE, Member IEEE

Pol-Mux QPSK signal
(PDM-QPSK)

4 bits/symbol

optical PDM
QPSK signal

1x25 Gbaud
1x100 Gbit/s

Fiber optic communication systems: limit of capacity ?

Fiber optic communication systems: limit of capacity ?

Capacity limit

How much information can we transmit in a given bandwidth?

⇒ Shannon theorem

Additive white Gaussian noise channel

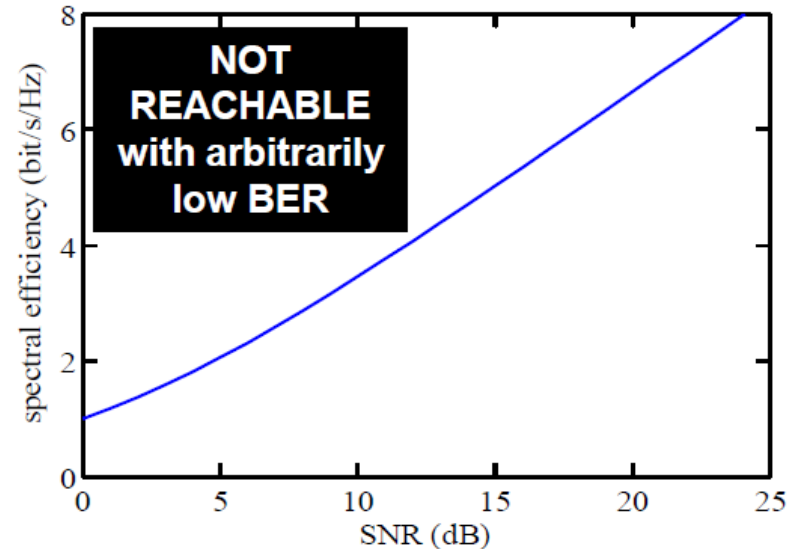
$$C = W \log_2(1 + \text{SNR})$$

channel
capacity

available
bandwidth

signal-to-noise
ratio

$$\text{SE}_{\max} = \frac{C}{W} \quad \text{maximum spectral efficiency (in bit/s/Hz)}$$



The Bell System Technical Journal

Vol. XXVII

July, 1948

No. 3

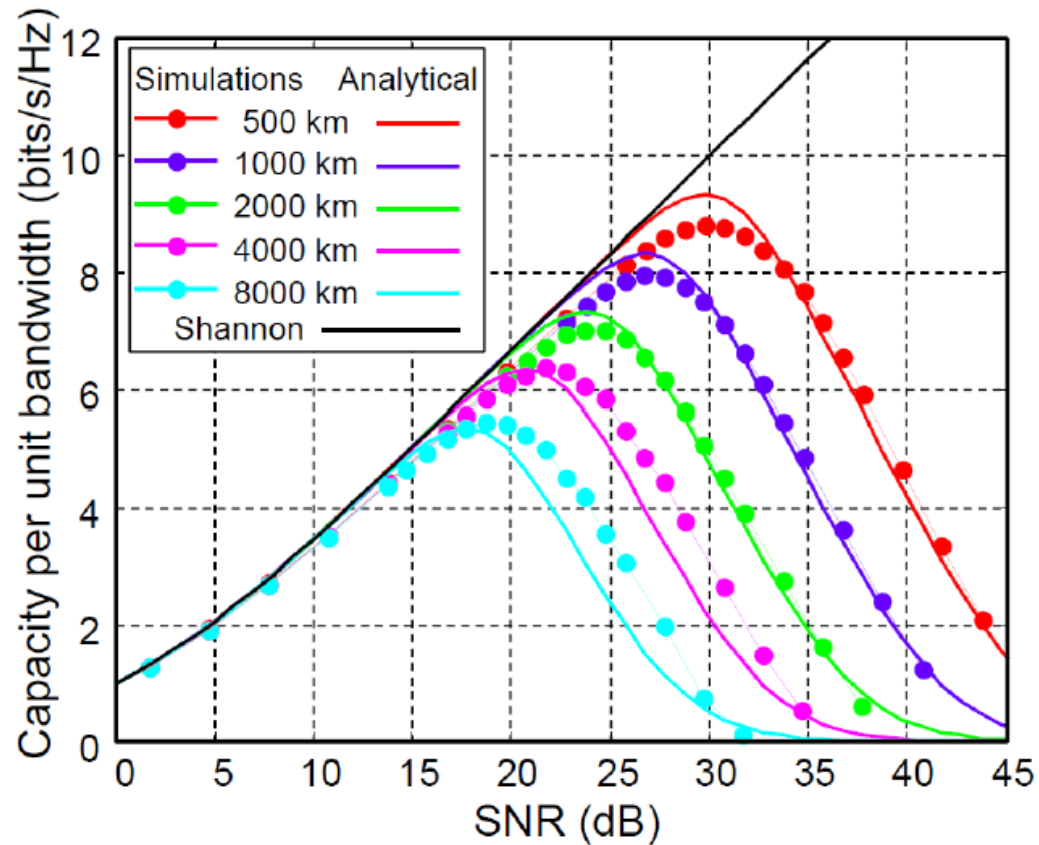
A Mathematical Theory of Communication

By C. E. SHANNON



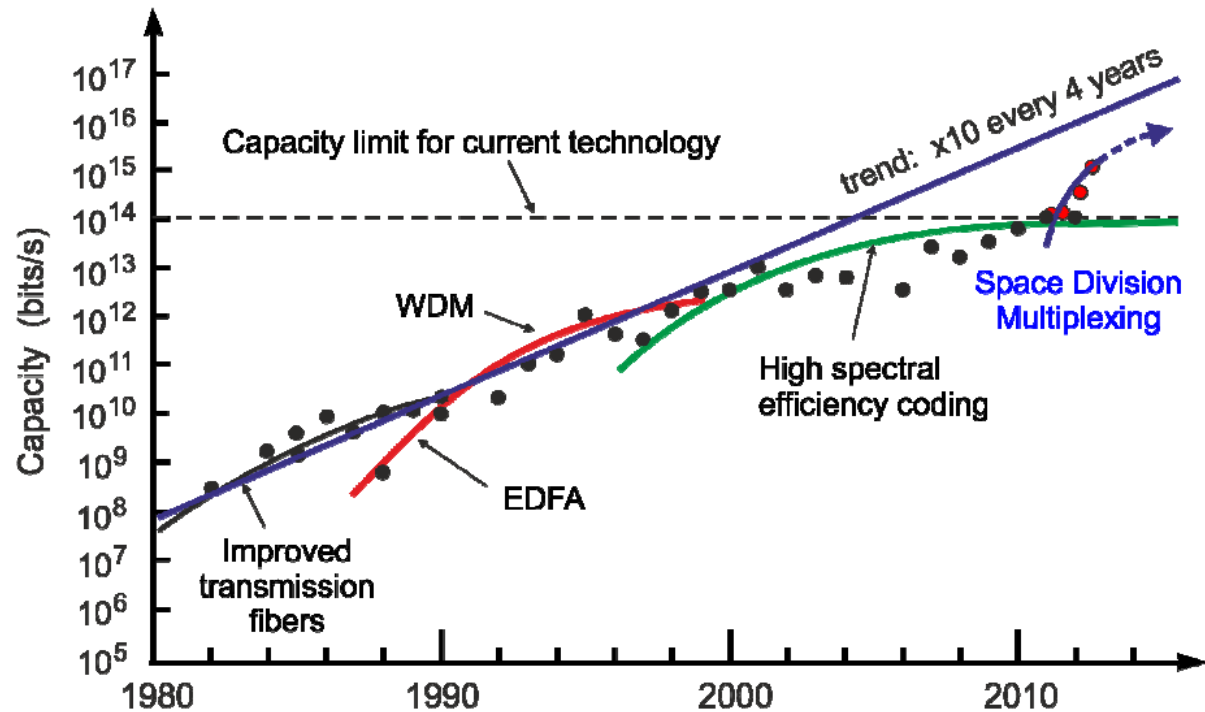
Claude Shannon
(1916-2001)

Capacity limits in optical fibers



Fiber optic communication systems: limit of capacity ?

How to further increase the transmission capacity?

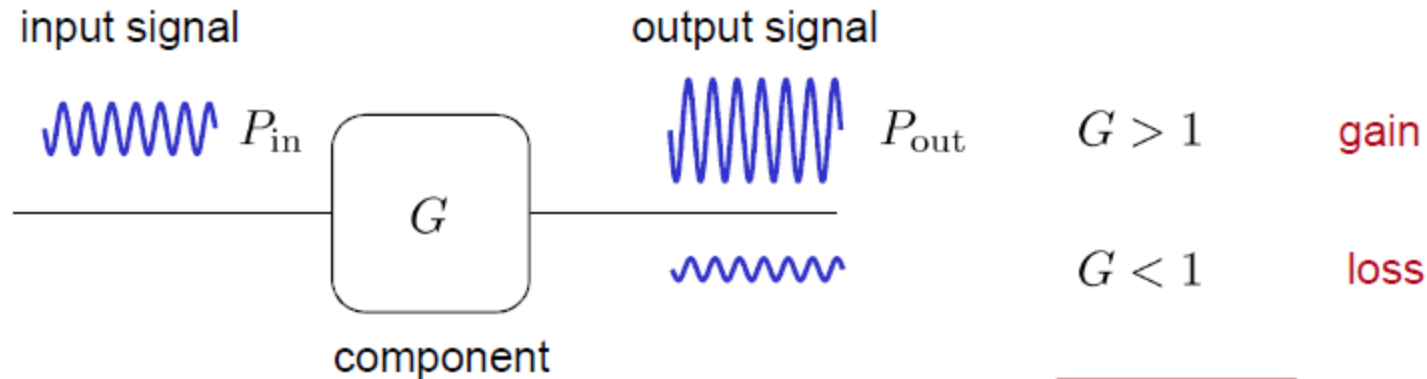


Space Division Multiplexing in Optical Fibres

D. J. Richardson, J. M. Fini, L E. Nelson

Fiber optic system: how to calculate the link power budget ?

The dB scale



The “gain” of the component is defined as:

$$G = \frac{P_{out}}{P_{in}}$$

In practice, the amplitude of the gain / loss gain values can be extremely large (loss / gain range within 0.001 to 1000 are not rare)

Convenient to express the gain in **logarithmic scale**, also known as **decibel** (dB)

$$G_{[dB]} = 10 \log_{10} \left(\frac{P_{out}}{P_{in}} \right)$$

$$G = 1 \quad G_{[dB]} = 0 \text{ dB}$$

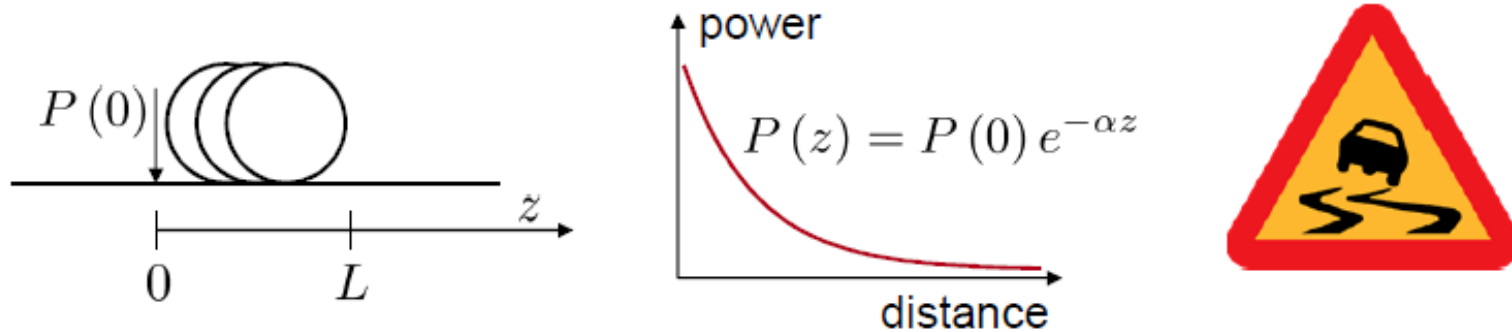
$$G = 10 \quad G_{[dB]} = 10 \text{ dB}$$

$$G = 100 \quad G_{[dB]} = 20 \text{ dB}$$

$$G = 0.01 \quad G_{[dB]} = -20 \text{ dB}$$

$$G = 2 \quad G_{[dB]} \simeq 3 \text{ dB}$$

Fibre attenuation: dB and linear units



Fibre loss usually expressed in units of dB/km

e.g. typically 0.2dB/km for standard transmission fibres

$\Rightarrow$ this value does not correspond to α in $P(z) = P(0) e^{-\alpha z}$: α in m^{-1}

Fibre “gain”:

$$G = 10 \log_{10} \left(\frac{P(L)}{P(0)} \right) = -0.2 \text{ dB} \quad \text{for} \quad L = 1000 \text{ m}$$

$$P(L) = P(0) e^{-\alpha L} \Rightarrow G = 10 \log_{10} (e^{-\alpha L}) = -\frac{10000}{\ln 10} \alpha \approx -4343 \alpha$$

Powers expressed in logarithmic scale

Until now we have seen that the logarithmic scale can be used to express **power ratios**:

$$G = \frac{P_{\text{out}}}{P_{\text{in}}} \quad \Rightarrow \quad G_{[\text{dB}]} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right)$$

One can also use it to **express power values with respect to a reference power**:

Example: $P_{[\text{dBW}]} = 10 \log_{10} \left(\frac{P}{1 \text{ W}} \right)$

In optical communication systems, the order of magnitude of powers is typically the **mW** and we will express powers in **dBm**

$$P_{[\text{dBm}]} = 10 \log_{10} \left(\frac{P}{1 \text{ mW}} \right)$$

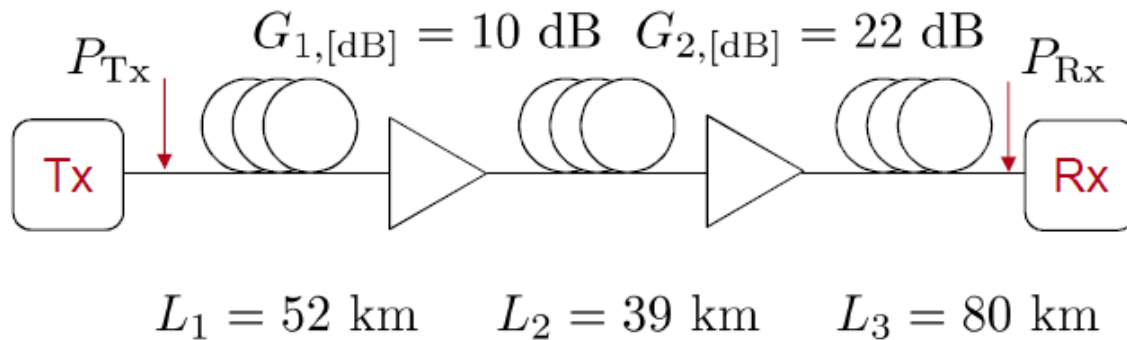
“power relative to 1 mW”

$$1 \text{ mW} \quad \Leftrightarrow \quad 0 \text{ dBm}$$

$$1 \text{ W} \quad \Leftrightarrow \quad 30 \text{ dBm}$$

$$1 \text{ }\mu\text{W} \quad \Leftrightarrow \quad -30 \text{ dBm}$$

The dB scale: the beauty of the system



Fibre loss:



$$P_{out} = P_{in} e^{-\alpha L}$$

$$\alpha = 4.6 \times 10^{-5} \text{ m}^{-1}$$

$$\alpha_{[dB/km]} = 0.2 \text{ dB/km}$$

Given the transmitter power P_{Tx} , calculate the received power P_{Rx}

In linear units:

$$P_{Rx} = P_{Tx} e^{-\alpha L_1} G_1 e^{-\alpha L_2} G_2 e^{-\alpha L_3} \quad (\text{product})$$

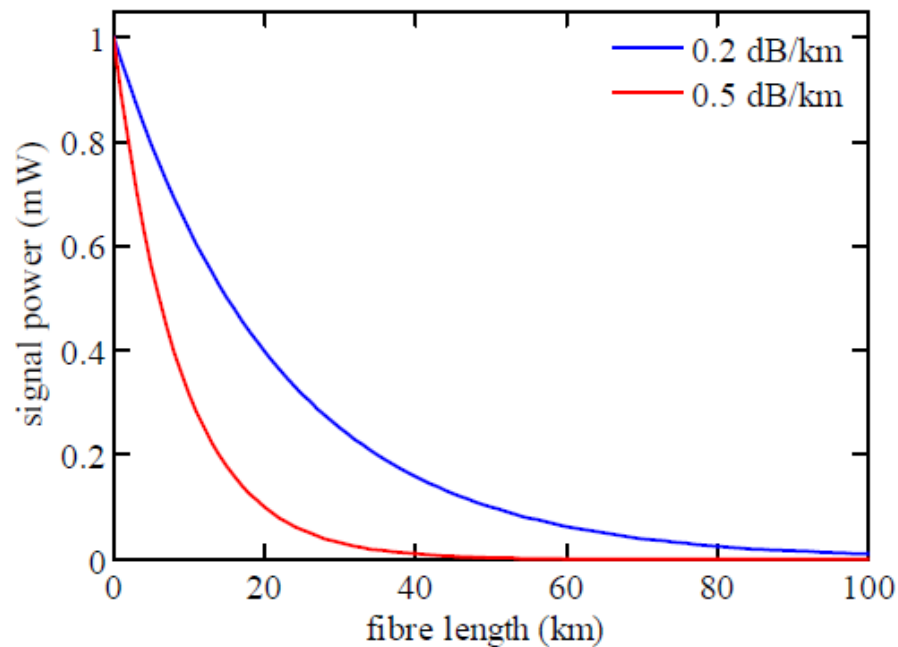
Or rather in dB units:

$$P_{Rx,[dBm]} = P_{Tx,[dBm]} - 0.2 \times 52 + 10 - 0.2 \times 39 + 22 - 0.2 \times 80 \quad (\text{sum})$$

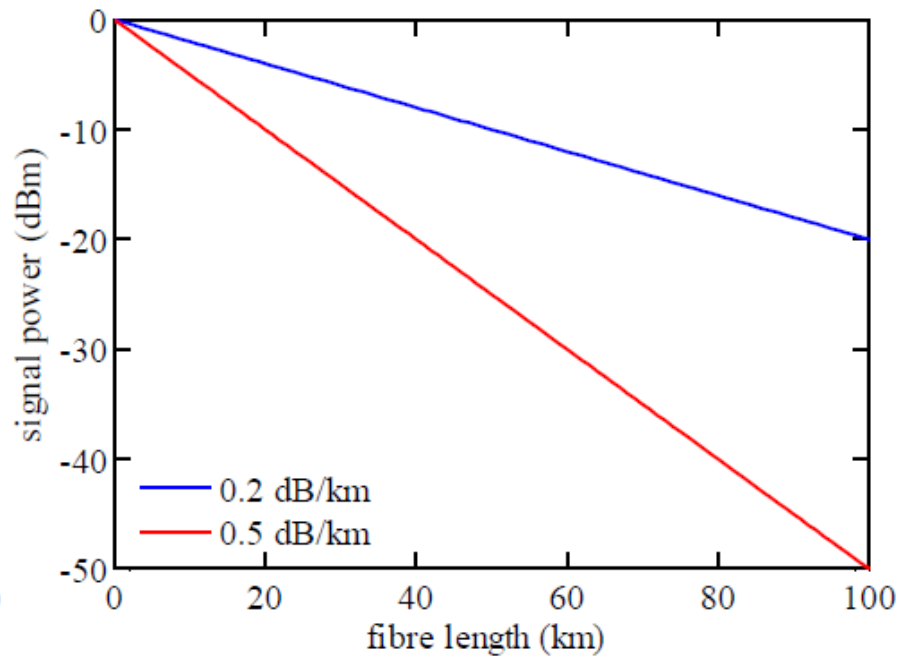
gain: “ + ” loss: “ - ”

Fibre loss and dB scale

Linear scale



Logarithmic scale



$$P_{\text{out}} = P_{\text{in}} e^{-\alpha L}$$

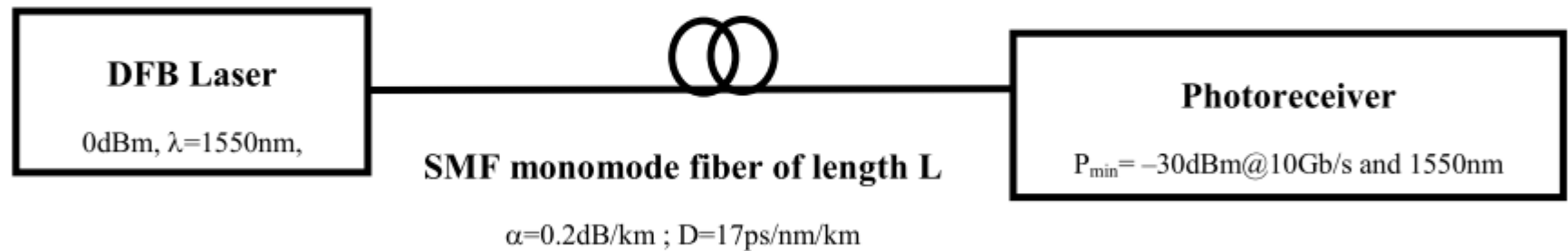
$$\alpha = 4.6 \times 10^{-5} \text{ m}^{-1}$$

$$P_{\text{out},[\text{dBm}]} = P_{\text{in},[\text{dBm}]} - \alpha_{[\text{dB/km}]} L_{[\text{km}]}$$

$$\alpha_{[\text{dB/km}]} = 0.2 \text{ dB/km}$$

Exercise 1: Introduction to optical amplification and the appropriateness of using it

We want to analyse the optical link depicted in the following figure. This link is composed of a laser delivering 0dBm @1550nm with a spectral width of 12.5GHz, a SMF fiber which length is L with losses of 0.2dB/km and a dispersion coefficient of 17ps/nm/km and a photoreceiver with a sensitivity of -30dBm. **We want to operate the optical link @10Gb/s.** We remind that the B.L (Bit rate.Length) is limited by $1/(|D|. \Delta\lambda)$.



- Calculate the limitation of the length L_{MAX1} for this optical link due to the losses.
- Calculate the limitation of the length L_{MAX2} for this optical link due to the dispersion.
- Compare L_{MAX1} and L_{MAX2} . Conclusion?